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Unsaturated bond recognition leads to biased signal in a fatty acid receptor

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Individual free fatty acids (FAs) play important roles in metabolic homeostasis, many via engagement with more than 40 GPCRs. Searching for receptors to sense beneficial ω-3 FAs of fish oil enabled the identification of GPR120, involving with a spectrum of metabolic diseases. Here, we report six cryo-EM structures of GPR120 in complex with FA hormones or TUG891 and G_i or G_{iq} trimers. Aromatic residues inside the GPR120 ligand pocket were responsible for recognizing different double-bond positions of these FAs and connect ligand recognition to distinct effector coupling. We also investigated synthetic ligand selectivity and the structural basis of missense single nucleotide polymorphisms. We reveal how GPR120 differentiates rigid double bonds and flexible single bonds and may facilitate rational drug design targeting to GPR120.

Individual free fatty acids (FAs) are important signaling molecules that modulate body weight, insulin secretion and sensitivity, inflammation, and many other metabolic processes (*1*–*3*), in addition to being critical energy sources. Omega-3 FAs (ω-3 FAs) in particular have well documented beneficial effects, including anti-inflammatory effects (*4*), preventing metabolic disorder (*5*), and enabling homeostasis of healthy fat tissue (*6*), and as do alterations in the unsaturated-saturated FA ratio using Fat1 transgenic mice (*3*, *4*). Searching for receptors mediating the anti-inflammatory effects of ω-3 FAs enabled the identification of GPR120 (FFAR4, free fatty acid receptor 4) (*7*, *8*), a receptor that can sense long-chain FAs (LCFAs) and promote insulin sensitization (*9*, *10*), anti-inflammatory effects (*5*, *11*), GLP-1 secretion stimulation (*11*, *12*) and the control of adipogenesis (*6*). Further research using synthetic GPR120 agonists and GPR120-deficient mouse models, as well as the manifestation of the association of the p.R270H missense mutation of GPR120 in obesity (*13*) supported the therapeutic potential of GPR120 for the treatment of metabolic diseases (*2*, *14*–*18*).

The activation of GPR120 is known to be coupled to multiple downstream effectors, including G_s, G_i, G_q and β-arrestins (*6*, *8*). While G_q-mediated signaling is required for GPR120-mediated GLUT4 translocation in adipose tissues and GLP-1 secretion (*7*, *12*), the G_i function of GPR120 plays critical roles in promoting insulin secretion in pancreatic islets and inhibiting ghrelin secretion in stomach cells (*19*, *20*). Moreover, mechanistic studies revealed that the anti-inflammatory effects of GPR120 are mediated by its association with β-arrestin2, which blocks the activities of TAK1 and the NLRP3 inflammasome (*5*, *7*). In addition, the G_s signaling of GPR120 at cilia potentially controls adipogenesis (Fig. 1A) (*6*). Although both saturated and unsaturated FAs were able to activate GPR120 and only certain unsaturated FAs are known to be beneficial for metabolism (*7*, *11*), the signaling bias of GPR120 in response to different saturated/unsaturated FAs, as well as synthetic compounds, has just begun to be appreciated by our group and others (Fig. 1A) (*16*).

In the present study, we set out to delineate the bias properties of GPR120 in response to stimulation with endogenous saturated 9-hydroxystearic acid (9-HSA), unsaturated linoleic acid (LA) and oleic acid (OA), the natural agonist ω-3 eicosapentaenoic acid (EPA), and the synthetic agonist TUG891 (*18*, *21*). We determined six GPR120 structures in complex with these agonists and G_i/G_{iq} transducers, which, along with pharmacological profiling, mutations and computational work, provided in-depth insights into double-bond recognition and signaling bias by GPR120.

Bias property of GPR120 agonists

Different downstream effectors play key roles in mediating distinct GPR120 (FFAR4) functions. We found previously that two endogenous FAs, OA and LA, were up-regulated in pancreatic islets in response to glucose and exhibited functional and signaling bias in the activation of mouse GPR120 (*16*), but the bias properties of different endogenous or natural FAs toward human GPR120 have not been characterized. We exploited arrestin recruitment and G protein dissociation assays to profile the bias properties of GPR120 toward a panel of agonists, including the ω-6 FA LA and the ω-9 FA OA (two endogenous agonists); the natural product ω-3 FA, EPA (a component of fish oil); and the synthetic agonist TUG891, using the saturated FA 9-HSA as a reference (*22*) (Fig. 1, B to F; fig. S1, A to E; and table S1). By direct comparison of the representative concentration-dependent curves of GPR120 ligands, bias differences could be identified in selective ligand sets and particular signaling pairs. For instance, 9-HSA, EPA, and TUG891 showed similar Emax and EC50 values in G_i signaling, but the G_q activities of TUG891 were higher than those of 9-HSA and EPA (Fig. 1, C to E; fig. S1, A to E; and table S1). We next used a two-dimensional differential σ plot to enable full visualization of the bias property between the G_q and G_i signaling pair. A positive σ value indicated stronger activity of the agonist than the reference compound in the selective signaling pathway (X axis or Y axis). A ligand over the diagonal line indicated bias toward a particular type of signaling (Y axis signaling) over the other type of signaling (X axis signaling), whereas a ligand below the diagonal line suggested a bias toward the X-axis. Importantly, TUG891 is a G_q bias ligand, whereas EPA showed weak G_i bias when we compared G_q activation over G_i using 9-HSA as a reference (Fig. 1F, fig. S2, and table S2) (*22*–*24*). In particular, only ω-3 FAs, including EPA, ALA and DHA, were able to activate G_s signaling, suggesting a strong G_s vs. G_i/G_q bias property of EPA compared with TUG891 or 9-HSA (Fig. 1E, fig. S1E, and table S1).

Human GPR120 has two alternative splicing forms, with one long (L) GPR120 splice variant (hGPR120-L) containing a 16 amino acid insertion in the third intracellular loop of the short (S) human GPR120 splice variant (hGPR120-S) (fig. S3A). hGPR120-L showed no detectable G_s or G_q activity in response to EPA, TUG891 or LA and much less G_i activity than hGPR120-S, possibly due to steric hindrance of ICL3 precluding G_i or G_q protein coupling (Fig. 1, G and H; figs. S1, F to N, and S3; and tables S1 and S3).

Structural determination and overall structure

We then solved the cryogenic electron microscopy (cryo-EM) structures of the TUG891-, 9-HSA-, LA-, EPA- and OA-bound GPR120-G_i complexes and the TUG891-bound GPR120-G_{iq} complex at 3.0 Å, 3.1 Å, 3.1 Å, 2.8 Å, 2.8 Å, and 2.6 Å, respectively (figs. S4 to S9 and table S5). The cryo-EM densities allowed assignment of the bound ligands and most residues of the GPR120 receptor, G_i or G_q heterotrimer and scFv16 (Fig. 2, A to C, and figs. S4H, S5F, S6F, S7G, S8G, and S9G). The binding modes of the ligands in the corresponding structures were further supported by molecular dynamics simulations (fig. S10).

GPR120 assumed a classic seven transmembrane (7TM) bundle topology (Fig. 2, A and B). Compared to other FA or lipid GPCRs, including GPR40 (PDB: 5TZR) (*25*), LPA6 (PDB: 5XSZ) (*26*) and S1PR1 (PDB: 3V2Y) (*27*), GPR120 showed striking differences at its N terminus and ECL2 (table S6). Whereas the N terminus of LPA6 or GPR40 was in proximity to ECL3, the N terminus of GPR120 assumed a “U” configuration lying across the 7TM bundle and inserted under ECL2, creating one side of the upper barrier of the ligand pocket (figs. S11, A and B, and S12A). ECL2 contained two β sheets and covered the N terminus of GPR120 (fig. S11A). In contrast, the N terminus of S1PR1 assumed an α-helical configuration and covered the ligand-binding pocket by juxtaposition with ECL2 (fig. S13A). The GPR120 ligand pocket was deeper than that of MK-8666-bound GPR40, and all agonists were buried deep inside the 7TM bundle (Fig. 2D and fig. S11, C and D). Whereas the ligand pockets of GPR120 were sealed, the GPR40 agonist MK-8666 was inserted diagonally into the receptor from an entrance created between TM3 and TM4 on the receptor side (Fig. 2D and fig. S12, B and C). For other comparisons, a proposed ligand pocket was formed between TM4 and TM5 of LPA6 to accommodate the main body of the LPA with an open entrance at the top (fig. S11, E and F) (*26*). Whereas the polar group of S1P was closer to TM1 and TM2, the polar groups of FAs were near TM5 in GPR120 (fig. S13, B and C). Compared with the LPA6, GPR40, and S1PR1 structures on the intracellular sides, GPR120 had a longer TM5 and TM6 to hold ICL3, and TM6 was shifted outward (figs. S11G, S12D, and S13D).

Fatty acid-binding pockets

Inside the ligand pocket, all FAs assumed an overall “L” configuration (Fig. 2, C and D). The convex side of these FAs is accommodated by a hydrophobic surface created by F27$^{\text{N-term}}$, F115$^{3.29}$, M118$^{3.32}$, L196$^{\text{ECL2}}$, I287$^{6.58}$ and F303$^{7.35}$. In contrast, the concave surfaces of the FAs faced a combination of hydrophobic and polar residues of TM3-TM4-TM5-TM6, including T119$^{3.33}$, L173$^{4.60}$, W207$^{5.38}$, F211$^{5.42}$ and I284$^{6.55}$. The hydrophobic aryl chain ends of these FAs were in contact with I281$^{6.52}$ and reached the bottom of the pocket formed by I126$^{3.40}$ and the toggle switch W277$^{6.48}$ (Fig. 2, E to I). Thirteen hydrophobic residues along the GPR120 ligand pocket contributed to interactions with all these FAs, thus potentially representing a common mechanism of recognition of FAs by GPR120 (Fig. 2, E to I; figs. S14, B and C, and S15; and table S7). The carboxylate termini of both 9-HSA and OA formed polar contacts with the main chain carbonyl of I284$^{6.55}$ and the main chain amine of L288$^{6.59}$ (fig. S16, A and B). In contrast, the carboxylate terminus of LA formed a water-mediated H-bond network with W198$^{\text{ECL2}}$, E204$^{5.35}$ and N291$^{\text{ECL3}}$, that of EPA formed only a hydrogen bond with the main chain carbonyl of I284$^{6.55}$ (Fig. 2G and fig. S16C). Consistent with these observations, mutation of the F115$^{3.29}$, I126$^{3.40}$, L173$^{4.60}$ or L196$^{\text{ECL2}}$ hydrophobic residue to alanine (Ala) decreased the G_i coupling activity of GPR120 in response to LA, OA, 9-HSA or EPA. Although the W198$^{\text{ECL2}}$A and E204$^{5.35}$A mutations had no effects on 9-HSA-induced GPR120 activation, disrupting the H-bond network between LA and GPR120 via either of these two residues impaired LA-induced GPR120 activation (fig. S16, D to K, and table S8). The hydrophobic corridor formed by 14 common hydrophobic amino acids may thus serve as a general mechanism for the recognition of FAs by GPR120, whereas the polar networks may contribute to FA recognition specificity.

Recognition of double bonds of unsaturated FAs by GPR120

9-HSA is a saturated FA, while LA, OA and EPA are unsaturated FAs and have double bonds in specific locations along carbon chains (Fig. 1B). Inspection of the structures of LA-, OA- and EPA-bound GPR120 offered detailed information on the interaction modes of the pocket residues with these unsaturated double bonds. Comparisons of the mutation effects of residues in contact with unsaturated double bonds between unsaturated FA-induced GPR120 activation and 9-HSA-induced GPR120 activation also provided us with an opportunity to define the contributions of pocket residues to specific double-bond recognition because the corresponding double bonds in unsaturated FAs were replaced by carbon–carbon single bonds in 9-HSA.

Notably, each double bond in OA (single double bond), LA (two double bonds) and EPA (five double bonds) interacts with 3-6 hydrophobic residues of GPR120 (Fig. 3, A to K). We systematically mutated contacting residues and inspected whether they contributed to double-bond recognition by comparing their effects on GPR120 activation induced by unsaturated FAs with those induced by 9-HSA. Mutations of aromatic residues led to decreased potency of unsaturated FAs compared with saturated FAs via recognition of these double bonds. For instance, the mutations F211$^{5.42}$A or W207$^{5.38}$A specifically decreased EPA-induced GPR120 activity more than the activity induced by 9-HSA and other unsaturated FAs. These mutational effects are consistent with specific interactions between W207$^{5.38}$ and F211$^{5.42}$ and the double bonds xv and iii of EPA, respectively. The mutation F88$^{2.53}$V led to a larger decrease in the activity of LA and EPA, which all had a double bond at the vi position, than in the activity of 9-HSA and OA (Fig. 3, L and M; fig. S17, A to H; and table S10). Specifically, F27$^{\text{N-term}}$, F115$^{3.29}$ and F303$^{7.35}$ constituted the central aromatic core to accommodate the aliphatic chain of the FAs, forming contacts with bond ix to xv in EPA and bond ix in OA and LA. Consistent with these observations, upon the mutation F27$^{\text{N-term}}$V, F115$^{3.29}$A or F303$^{7.35}$V, the potency of EPA, LA and OA decreased more than that of saturated 9-HSA (Fig. 3, L and M; fig. S17, A to H; and table S10). In contrast, the nonaromatic hydrophobic residues seem to prefer single bonds over double bonds; for example, the activity of M118$^{3.32}$A decreased approximately 50-, 5-, 4- and 2-fold in response to stimulation with 9-HSA, OA, LA and EPA, respectively. I280$^{6.51}$A led to approximately 25-, 10-, 2- and 2-fold decreases in the potency of 9-HSA, OA, LA and EPA with regard to GPR120, respectively (Fig. 3L; fig. S17, A to G; and table S10). We speculated that these effects were due to the greater flexibility of the single-bond/nonaromatic hydrophobic side chain than the double-bond/aromatic side chain, enabling easier fitting of the single bond to the flexible conformations of nonaromatic residues for favorable interactions.

To further investigate the mechanisms of double-bond recognition by GPR120, we performed molecular dynamics simulations. The π:π interactions between aromatic residues and double bonds are known to contribute to intermolecular interactions in organic compounds or proteins (*28*, *29*). We performed long-term molecular dynamics simulations of the LA-GPR120, OA-GPR120, and EPA-GPR120 complex systems using constraint algorithms and simulated annealing by gromacs (fig. S18). The software's own gmx distance module was used to analyze the distances between aromatic residues and double bonds in the simulation trajectories. Our calculations indicated that aromatic residues were more prone to forming stable contacts with double bonds via π:π interactions than their interactions with single bonds at the same position (Fig. 3, N and O, and fig. S17, I to N). Moreover, the mutations of the corresponding aromatic residues to nonaromatic Leu residues also significantly decreased their stable interactions with the double bonds (Fig. 3P and fig. S19).

Collectively, our studies indicated that the π:π interactions contributed by aromatic residue arrays located in ligand pockets may be an important mechanism for selective double-bond recognition by GPR120. In particular, each of these aromatic residue arrays, such as $F27^{N\text{-term}}$, $F88^{2.53}$, $F115^{3.29}$, $W198^{ECL2}$, $W207^{5.38}$, $F211^{5.42}$, and $F303^{7.35}$, recognizes one to three double bonds at specific positions of these unsaturated FAs.

Recognition of synthetic TUG891 and ligand pocket selectivity of GPR120

Selective activation of GPR120 has therapeutic potential for the treatment of many diseases, including type 2 diabetes mellitus, osteoporosis, inflammation and obesity (*1*, *2*, *7–9*, *11*, *13–15*, *30*). The GPR120-selective agonist TUG891 has been widely used in functional studies of GPR120 (*18*, *21*). To understand the structural basis of the ligand selectivity of GPR120 and facilitate the future design of selective GPR120 agonists, we determined the cryo-EM structures of TUG891-GPR120-G_i and TUG891-GPR120-G_{iq} complexes (figs. S5 and S9). The EM density for the O10 of TUG891 is lacking in the TUG891-GPR120-G_i complex structure, whereas the EM density for TUG891 in the TUG891-GPR120-G_{iq} complex is continuous, and the modeled compound fits well with the EM density (Fig. 2, B and C, and fig. S20, A to C). The binding pose of TUG891 in the ligand pocket of GPR120 was further confirmed by molecular dynamics (MD) simulation (fig. S10A). This better EM map for TUG891 in the GPR120-G_{iq} complex than the EM map in the GPR120-Gi complex may be consistent with TUG891 having better G_q activity than G_i (TUG891 may have a more stable conformation in the GPR120-G_{iq} complex) (Fig. 1, C, D, and F). Therefore, we mainly used the TUG891-GPR120-G_{iq} structure to describe the interaction between TUG891 and GPR120 and performed computational simulation of the potential position of TUG891 in the TUG891-GPR120-G_i complex structure (Fig. 4A and fig. S10E).

The presence of three aromatic rings in TUG891 rendered the compound more rigid than endogenous FA agonists. Divided by oxygen O10, the compound was separated into two parts, the phenylpropionic acid in the upper region (region A) and the 4-fluoro-phenyl ring and the 4-methyl phenyl ring in the lower region (region B) (Fig. 4, A and B). Notably, TUG891 is a selective GPR120 agonist. Sequence alignment indicated that at least five residues of TM2 and TM5-TM6, including $F88^{2.53}$, $W207^{5.38}$, $F211^{5.42}$, $N215^{5.46}$ and $I287^{6.58}$, are distinct compared to other GPCRs known to sense LCFAs (Fig. 4C). Mutation of these residues in GPR120 to allelic residues in GPR40, FFAR2, FFAR3, and GPR119 receptors, including $F88^{2.53}$A/L, $W207^{5.38}$A/V, and $I287^{6.58}$ A/S, $F211^{5.42}$A/V and $N215^{5.46}$A/L all decreased GPR120 activity in response to TUG891 stimulation (Fig. 4, C and D; fig. S21; and table S12). It may thus be possible to target these pocket residues for selective GPR120 agonist design. Moreover, the separation of TUG891 into two segments occupying different hydrophobic pockets by the O10 linker suggested that fragment-based chemical synthesis could be exploited to develop selective GPR120 agonists (Fig. 4, B and D; fig. S21; and table S12).

Contribution of pocket residues to signaling bias of GPR120

The diverse bias patterns of endogenous and synthetic agonists made GPR120 a good example for investigation of the structural and molecular mechanisms underlying the signaling bias of an FA GPCR (Fig. 1). TUG891 has three aromatic rings, which were found to form extensive π:π interactions with aromatic residues of GPR120 and to show significant G_q bias compared with EPA and 9-HSA (Fig. 1, C, D, and F). We therefore profiled effects of mutations of residues surrounding TUG891 on G_q and G_i activation downstream of GPR120. The mutations $F27^{N\text{-term}}$A, $F28^{N\text{-term}}$L, $F88^{2.53}$A, $F115^{3.29}$A, $F177^{4.64}$A, $F216^{5.47}$A, $F303^{7.35}$A and $F311^{7.43}$A induced much more loss of G_q activation by TUG891 than of G_i activation (Fig. 5, A and B; fig. S22, A to C; and table S13). These results suggested that interactions of TUG891 with an array of eight aromatic residues in the GPR120 ligand pocket may serve as an important mechanism for the G_q bias activity of TUG891, and this aromatic residue array could be exploited for design of G_q biased agonists targeting GPR120 (Fig. 5, A and B; fig. S22, A to C; and table S13).

In particular, only the ω-3 FAs, but not other FAs among our tested FAs or synthetic TUG891, showed G_s activity, indicating a strong G_s bias of ω-3 FAs compared with other tested FAs. Compared with 9-HSA, a saturated FA with similar G_i activation, EPA formed multiple additional π:π interactions with $F27^{N\text{-term}}$, $F115^{5.39}$, $W198^{5.38}$, $W207^{5.38}$, $F211^{5.42}$ and $F303^{7.35}$ via its double bonds along the carbon chain (Fig. 3, G to J). Mutations of $F88^{2.53}$A, $F211^{5.42}$L and $F303^{7.35}$A in particular showed larger effects on the EPA-induced G_s activation of GPR120 than on G_i activation, suggesting that the recognition of double bonds iii, vi and xii by these three residues determines the G_s bias property (Fig. 5, C and D; fig. S22, D to G; and table S14). In addition, $I287^{6.58}$ formed stronger hydrophobic interactions with EPA, and the $I287^{6.58}$A mutation caused a larger loss of EPA-induced G_s activation than G_i activation (Fig. 5D). Therefore, not only π:π interactions mediated by the double bond decorations of unsaturated FAs but also other hydrophobic interactions within the ligand pocket contribute to the specific G_s activation of EPA (fig. S22, E to G, and table S14).

Collectively, our data suggested that the π:π interactions between the unsaturated double bonds presented by the unsaturated FAs and the specific combinations of arrays of aryl amino acids located in pockets of GPR120 and other specific hydrophobic contacts played important roles in determining G_q vs. G_i and G_s vs. G_i signaling bias.

Active states of GPR120

In all six GPR120-G_i or GPR120-G_{iq} structures in complex with either endogenous or natural FAs, as well as the synthetic compound TUG891, the agonists were in direct contact with the toggle switch W277$^{6.48}$, which assumed similar configurations (fig. S23, A to G). Below W277$^{6.48}$, the triad connector P$^{5.50}$-I$^{3.40}$-F$^{6.44}$ motif formed tight packing (fig. S23, H to M), which has been proposed to be essential for maintaining the active state in many class A GPCRs (*23*, *31–33*). On the cytoplasmic side, separation of the cytoplasmic termini between TM3 and TM6, breakage of the ionic lock [constituted by the ERM motif (D/ERY motif in class A GPCRs) and D259$^{6.30}$], and packing of the N317$^{7.49}$P318$^{7.50}$XXY321$^{7.53}$ motif with R136$^{3.50}$ were observed (fig. S23, N to Y), which are all hallmarks of the active state of class A GPCRs.

Coupling between GPR120 and G_i protein

The overall interaction mode between G_i and GPR120 was mostly similar to the binding of G_i with the lipid receptor S1PR1; however, the detailed contact residues and the orientation of second structural elements still showed substantial differences (Fig. 6, A to D, and fig. S24, A to C). For example, the C-terminal tail of the α5 helix was rotated away from the TM3 and toward TM6 by approximately 8° in the 9-HSA-GPR120-G_i complex (Fig. 6A). The C-terminal wavy hook of G_i was inserted into the cavity formed by TM2-TM3 and TM5-TM7 of GPR120 (fig. S24A) and formed specific hydrophobic, polar and charge-charge interactions (figs. S24B and S25, A to F). Another notable feature of the GPR120-G_i interface is the involvement of all three intracellular loops (ICLs), composed of 8~11 residues in each ligand induced GPR120-G_i complexes, in all five GPR120-G_i complex structures and occupying approximately half the total G_i trimer-contacting amino acids of GPR120 (fig. S24C and table S15). In GPR120, there are five positively charged arginine residues located in ICL1, which is close to an acidic patch of D350$^{G.H5.22}$ and carboxylic end of F354$^{G.H5.26}$ of α5 helix of $G\alpha_i$ (Fig. 6B and figs. S25A and S26A). Although ICL3 of most class A GPCRs is disordered, ICL3 of GPR120 has a well-defined EM density and forms extensive interactions with the G_i protein (Fig. 6D and fig. S24C). Whereas the N terminus residues of ICL3 formed polar, charge or hydrophobic interactions mainly with the α5 helix of $G_{\alpha i}$, several positive charged or hydrophobic residues of the C terminus of ICL3 and initial helical of TM5 of GPR120 folded inside the 7TM bundles and formed contacts with the β-strand 6 and α4-α5 helix of $G_{\alpha i}$ (Fig. 6C and figs. S24D and S25, B and G). These observations were further supported by mutational analysis of G_i coupling activity of GPR120 in response to 9-HSA stimulation (Fig. 6I; fig. S27, A and B; and table S16).

Coupling between GPR120 and G_q

Inspection of the G protein interface of the TUG891-GPR120-G_{iq} complex revealed that 21 residues of GPR120 contacted with 15 residues of $G\alpha_{iq}$ (Fig. 6D). Among these $G\alpha_{iq}$ residues, 12 are identical in native $G\alpha_q$ and 3 residues are different (fig. S26B). We therefore substituted these 3 unidentical $G\alpha_{iq}$ interface residues with native $G\alpha_q$ residues and performed computational simulation (Fig. 6D and fig. S24, E and F). The simulation results revealed three additional GPR120 residues may contact with $G\alpha_q$, in addition to the other 21 residues identified in GPR120-G_{iq} interface (Fig. 6D). Further extensive mutagenesis analysis indicated at least 14 residues of GPR120 on the GPR120-G_q interface affected $G\alpha_q$ coupling in response to TUG891 stimulation (Fig. 6I; fig. S27, A and C; and table S16). Comparison of the TUG891-GPR120-G_{iq}, simulated TUG891-GPR120-G_q and 9-HSA-GPR120-G_i complex structures revealed both shared and different interactions (Fig. 6, D to H, and figs. S24, E to J, and S25, H to J). Importantly, 19 out of 24 residues that mediated both G_i and G_q coupling formed conserved interactions with αN and α5 of both $G\alpha_i$ and $G\alpha_q$, thus may accounting for the ability of GPR120 to induce both types of G protein signaling (Fig. 6, D and I; fig. S27, A to C; and tables S15 and S18). Notably, differences between GPR120-G_q interface and GPR120-G_i interface were also observed (Fig. 6, E to H, and fig. S24, G to J). For example, the replacement of D337$^{G.H5.09}$ and T340$^{G.H5.12}$ of $G\alpha_i$ by A342$^{G.H5.09}$ and K345$^{G.H5.12}$ of $G\alpha_{iq}$ introduced charge-charge repulsion and destroyed previous charge-charge and polar interactions formed between R240^{ICL3} of GPR120 with D337$^{G.H5.09}$ and T340$^{G.H5.12}$ of $G\alpha_i$ in the 9-HSA-GPR120-G_i complex (Fig. 6, E and F); the replacement of K345$^{G.H5.17}$ of $G\alpha_i$ by Q350$^{G.H5.17}$ of $G\alpha_{iq}$ eliminated its charge interaction with D259$^{6.30}$ but enabled the formation of a new hydrogen bond with R238^{ICL3} (Fig. 6, G and H, and fig. S25, C and D). Conversely, the replacement of G352$^{G.H5.24}$ and C351$^{G.H5.23}$ of $G\alpha_i$ by N357$^{G.H5.24}$ and Y356$^{G.H5.23}$ of $G\alpha_{iq}$ allowed new H-bond formation and hydrophobic packings in the TUG891-GPR120-G_{iq} complex structure, respectively (figs. S24, I and J, and S25, I and J). Consistent with these observations, the mutation of R240$^{5.71}$A, S248^{ICL3}A or D259$^{6.30}$A of GPR120 greatly impaired G_i but not G_q coupling in response to TUG891 stimulation (Fig. 6I; fig. S27, A to C; and table S16). In contrast, specific mutations of charged or hydrophobic residues of TM3 or TM5 of GPR120 impaired $G\alpha_q$ coupling more than $G\alpha_i$ coupling (Fig. 6I and fig. S27, A to C).

Interfaces between GPR120 and G_s characterized by MD simulation and mutagenesis

Among the 23 residues of GPR120 in contact with G_s in our simulated EPA-GPR120-$G\alpha_s$ complex structure, 15 residues also contacted $G\alpha_i$, suggesting that common interfaces of GPR120 are shared by these two different G protein subtypes

(Fig. 6D and fig. S28). Computational simulation and mutagenesis analysis also suggested different features of the GPR120-G_s interface compared with those of the GPR120-G_i interface (Fig. 6, D and J to N; figs. S25E and S27, A, B, and D, and S28; and table S16). For instance, the replacement of the α5 helical end residue C351 of $G\alpha_i$ by Y391 of $G\alpha_s$ allowed the formation of a potential H-bond with E135$^{3.49}$ and potential π:π interactions with Y227$^{5.58}$ in the simulated EPA-GPR120-$G\alpha_s$ structure (Fig. 6, K and L, and fig. S25). The replacement of I344 in the α5 helix of $G\alpha_i$ by Q384 from $G\alpha_s$ enabled the formation of a potential H-bond with Q144^{ICL2} (Fig. 6, M and N, and fig. S25E). Consistent with these predictions, the mutations E135$^{3.49}$A, Q144^{ICL2}A, and Y227$^{5.58}$A of GPR120 showed larger decreases in G_s activation than of G_i activation in response to EPA stimulation (Fig. 6J; fig. S27, A, B, and D; and table S16). Although the computational simulations and biochemical results allowed us to propose a preliminary model, the details of the interactions between GPR120 and G_q or between GPR120 and G_s require further structural characterization.

Potential propagation paths underlie differential G protein subtype coupling

In the TUG891-GPR120-G_q complex structure, the multiple π:π interactions of three phenyl rings of TUG891 with the aromatic array of five phenylalanines in the ligand pocket of GPR120 enabled combined conformational rearrangements of these residues compared with those in 9-HSA-GPR120-G_i complex structure (fig. S29A). In particular, the rotations of F88$^{2.53}$ and F311$^{7.43}$ connected to the conformational changes of L86$^{2.51}$ and I89$^{2.54}$, disrupting the H-bond formed between N313$^{7.45}$ and N317$^{7.49}$ and enabling constitution of the underneath polar network by residues N58$^{1.50}$, D85$^{2.50}$ and N317$^{7.49}$ (Fig. 7, A and B). These conformational changes were linked to structural rearrangements of hydrophobic packings starting from F82$^{2.47}$ and ending with R68^{ICL1} and V65$^{1.57}$, allowing outward tilting of TM7 and inward tilting of TM1 to enable a tighter insertion of the α5 helix of $G\alpha_q$ (Fig. 7C and fig. S29B). These structural changes propagated to a stronger hydrophobic packing between Y227$^{5.58}$, Y321$^{7.53}$, and R136$^{3.50}$, enabling the formation of cation-π interactions between R136$^{3.50}$ of GPR120 and Y356 of $G\alpha_q$ (Fig. 7, A and C). Consistent with these predictions, the G_q-biased property of TUG891 was reduced by mutations of propagating path connected to F88$^{2.53}$ and F311$^{7.43}$ in the ligand-binding pocket of GPR120 and the cytoplasmic side Y227$^{5.58}$ and Y321$^{7.53}$ (Fig. 7D; figs. S22, B and C, and S29, C to E; and tables S13 and S19).

Comparison of the 9-HSA-GPR120-G_i structure with our simulated EPA-GPR120-G_s model identified potential conformational locks connecting specific π:π interactions inside the ligand pocket to the cytoplasmic G protein coupling interface of GPR120, contributing to G protein subtype selectivity (Fig. 7E). The engagement with the ω-3 double bond enabled flipping of the aromatic ring of F211$^{5.42}$, accompanied by upward movements of the side chains of N215$^{5.46}$ and S123$^{3.37}$ in the MD model of EPA-GPR120-G_s (Fig. 7, E and F). These conformational changes were then propagated through a pathway connected by Y165$^{4.52}$-L127$^{3.41}$-I162$^{4.49}$-L158$^{4.45}$-L77$^{2.42}$ and finally led to structural rearrangement of the ionic lock residue of E135$^{3.49}$, which enabled hydrogen bonding with Y391 of G_s (Fig. 7, E to G). Consistent with this proposed model, the biased property of G_s is acutely sensitive to mutagenesis of the propagation path connected to F211$^{5.42}$ in the ligand-binding pocket of GPR120 and the cytoplasmic side E135$^{3.49}$ (Fig. 7H; fig. S29, C, F, and G; and table S21).

Taken together, our studies suggested that the propagation path connecting the double bond recognition of GPR120 inside the ligand pocket to specific structural features of the cytoplasmic side responsible for G_s/G_i signaling differentiation is mainly dependent on conformational locks located at TM3 and TM4. In contrast, the conformational locks at TM1-TM2 and TM7 linked the specific π:π interactions within the ligand pocket of GPR120 to cytoplasmic structural rearrangement for selective G_q vs. G_i coupling.

Disease-associated SNP of GPR120

It has been reported that the naturally occurring single nucleotide polymorphism (SNP) p.R270H, rs116454156 resulted in a missense mutation of Arg270 in TM6 of GPR120, which was coupled with increased fasting glucose levels and decreased homeostasis of pancreatic β-cells (*13*, *34*). Previous biochemical analyses have shown that p.R270H of GPR120 decreased G_q activity in response to α-linolenic acid (ALA) stimulation. The side chain of Arg254$^{6.25}$ of the GPR120 short isoform (the allele residue of p.R270H in the GPR120-long isoform) (Fig. 7I) directly interacts with the side chains of E308 and the main chain carboxyl group of K314 of $G\alpha_i$ via charge-charge or polar interactions in the LA-GRP120-G_i complex (Fig. 7J and fig. S25K). Similar interactions were observed in all other GPR120-G_i complex structures bound to different ligands (figs. S30, A to D, and S25, L to O). The computational simulation suggested that the replacement of Arg254$^{6.25}$ with His abolished the charge-charge interactions formed between R254$^{6.25}$ and E308 of $G\alpha_i$ (fig. S30, E and F). The mutation R254H reduced G_i coupling of GPR120 in response to either LA or EPA stimulation (Fig. 7K; fig. S30, G to I; and table S23), in line with recent reports that p.R270H may disturb islet homeostasis (*34*). In our simulated TUG891-GPR120-G_q complex structure, R254 formed an H-bond with the main chain of S314 of $G\alpha_q$ (fig. S31A). Consistently, the G_q activity of GPR120 was decreased by more than 3-fold by the mutation R254H (Fig. 7K; fig. S30, H and L to N; and table S23). In addition to impaired G_i and G_q signaling, all three tested ω-3 FAs induced G_s-$G\gamma$ dissociation responses that were approximately 2-fold lower in HEK293 cells overexpressing the GPR120 R254H mutant than in cells overexpressing wild-type GPR120 (Fig. 7K; fig. S30, O to Q; and table S23), which may be involved in adipogenesis in response to stimulation by ω-3 FAs. This effect was consistent with our simulation results, which indicated that R254H may impair the H-bonding between R254 of GPR120 and T350 of $G\alpha_s$ (fig. S31B).

Discussion

FAs with unsaturated double C-C bonds, such as ω-3 unsaturated long FAs (UFLFAs), are able to promote beneficial cellular signaling to modulate metabolism via direct action on membrane receptors, including GPR120 (*3*, *5*–*7*). In this study, we identified an array of aromatic residues, that were able to individually recognize specific double C-C bonds present in unsaturated FAs. Further structural analysis, biochemical characterization and molecular dynamics simulation suggested that these aromatic residues mainly recognize double bonds through π:π interactions and maintain a relatively stable binding mode with defined angles and distances, thus providing a structural framework for how membrane receptors can sense unsaturated FAs decorated with double bonds at different positions. The recognition of double bonds of unsaturated FAs by individual aromatic residues of GPR120 was connected to selective downstream G protein activation or β-arrestin recruitment and modulated the bias properties of GPR120. We characterized the potential propagating paths connecting the π:π interactions located in the ligand pocket to selective cytoplasmic structural features to differentiate downstream G protein subtype coupling. Whereas the conformational locks in TM3 and TM4 played important roles in transducing the specific double bond recognition of GPR120 to cytoplasmic G_s coupling, the conformational locks in TM1-TM2 and TM7 contributed significantly to linking the ligand pocket to downstream G_q vs. G_i discrimination. We recently found that an endogenous FA-GPR120 circuit with biased signaling properties was present in pancreatic islets and that a β-arrestin2-biased GPR120 endogenous agonist was able to improve islet functions and glucose metabolism in diabetic db/db mice (*16*). Now we see that unsaturated endogenous FAs and natural products show bias compared to saturated FA 9-HSA. Our pharmacological profiling indicated that only fatty acids harboring the ω-3 double bond were able to exhibit G_s activities. This might be one of the key features of these ω-3 unsaturated FAs that contribute to their beneficial effects on metabolism, as suggested by recent studies (*6*). Structural analysis indicated that the specific recognition of the double bonds in FAs by corresponding aromatic residues, such as the recognition of the ω-3 double bond by $F211^{5.42}$, is a determinant for their biased signaling. We speculate that unsaturated FAs may contribute to their beneficial effects via action on specific aromatic residues of GPR120, which could be exploited for the development of selective GPR120 agonists. A complete structural understanding of the GPR120 bias properties will require the future determination of the complex structures of GPR120 with G_s and β-arrestin1/2 and even with downstream effectors of arrestins, such as TAB1 and NLRP3 (*5*, *7*).

In addition to revealing mechanisms underlying unsaturated FA differentiation and signaling bias, our structure of the TUG891-GPR120-G_i and TUG891-GPR120-G_{iq} complexes afforded a structural basis for the agonist selectivity of GPR120. Most of the TUG891-interacting residues showed great diversity compared to other known FA receptors. Biochemical analysis indicated at least five residues governing the agonist selectivity of GPR120. In particular, the linker oxygen O10 separated TUG891 into two segments fitting into two binding pockets. These observations indicate that separate optimization of the two fragments individually could be exploited for future synthetic agonist design for GPR120-targeted therapies.

Materials and methods

Cells

HEK293 cells were obtained from the Cell Resource Center of Shanghai Institute for Biological Sciences (Chinese Academy of Sciences). *Spodoptera frugiperda* (Sf9) insect cells were purchased from the Expression Systems (catalog number: 94-001S).

Construct

The cDNA of human GPR120 short isoform (GPR120-S, NP 001182684.1) and human GPR120 long isoform (GPR120-L, NP 859529.2) were synthesized at BGI (Beijing, China). For structural determination, the sequence encoding for human GPR120-S (hereafter refers as GPR120), residues from 1-361, was cloned into a modified pFastBac1 vector with a haemagglutinin (HA) signal sequence at the N terminus. To facilitate expression and purification, a FLAG tag (DYKDDDDA) and cytochrome b_{562}RIL (BRIL) (*35*) followed by a three amino acid (Gly-Ser-Ala) linker was inserted to the N-terminal of GPR120. The fusion of BRIL into the N terminus of GPR120 does not significantly affect the ability of ligands-induced activation of GPR120 downstream signaling (fig. S32). The human $G\alpha_{i1}$ used in GPR120-G_i complexes was described in previous studies (*36*). For GPR120-G_q complex, an engineered $G\alpha_{iq}$ construct was generated based on $G\alpha_{i1}$, in which the 26 residues on α5 helix of $G\alpha_{i1}$ (residues 329-354) were replaced by that of $G\alpha_q$ (residues 334-359). Human Gβ1 and Gγ2 were cloned into the pFastBac-Dual vector. It is well established that the C-terminal region of a Gα subunit is one of the main determinants of coupling specificity (*37*, *38*) and similar strategy (using G_{sq} chimera) has been extensively used by Bryan Roth's and Eric Xu's group in determination of series of structures of GPCR-G_q complexes, including 5HTR2A-Gq, MRGPRX2-Gq, B1R-Gq, B2R-Gq (*39–41*). Moreover, additional computational simulation for TUG891-GPR120-G_q complex using TUG891-GPR120-G_{iq} complex as a template were performed and further validated by extensive mutagenesis analysis. For cell-based functional assays in HEK293 cells, human GPR120-L or GPR120-S was cloned into pcDNA3.1, with an N-terminal HA signal peptide followed by a FLAG tag, resulting in pcDNA3.1-GPR120-L and pcDNA3.1-GPR120-S, respectively. For β-arrestin recruitment assay, a yellow fluorescent protein (YFP) tag was added to the C-terminal of corresponding GPR120 constructs. The GPR120 mutants were generated using the Quikchange mutagenesis kit (Stratagene). All constructs were verified by DNA sequencing.

Protein expression

The *Spodoptera frugiperda* (Sf9) insect cells were infected at a density 2.5×10^6 cells per ml, four separate baculoviruses (GPR120, $G\alpha_{i1}$ or $G\alpha_{iq}$, $G\beta1\gamma2$ and scFv16) were co-added at a ratio of 1:1:1:1 into insect cells (*36*). After culturing for 48 hours at 27°C, the cells were harvested by centrifugation, and cell pellets were stored at −80°C.

GPR120-G_i and GPR120-$G\alpha_{iq}$ complex formation and purification

For the purification of GPR120-G_i or GPR120-$G\alpha_{iq}$ complexes, cell pellets were resuspended in buffer containing 20 mM HEPES pH7.5, 100 mM NaCl, 10 mM $CaCl_2$ and 5 mM $MgCl_2$, supplemented with complete Protease Inhibitor Cocktail tablets (Roche). Complex formation was initiated by adding 10 μM of TUG891 and apyrase (25 mU/ml, NEB) for both GPR120-G_i and GPR120-$G\alpha_{iq}$ complexes, or by adding 10 μM 9-HSA or 50 μM LA or 50 μM OA or 50 μM EPA and apyrase (25 mU/ml, NEB) for GPR120-G_i complexes. The suspension was incubated for 1 h at room temperature and then solubilized by 0.5% (w/v) lauryl maltose neopentyl glycol (LMNG, Anatrace) supplemented with 0.05% (w/v) cholesteryl hemisuccinate (CHS) (Anatrace) for 2 h at 4°C. Insoluble material was removed by centrifugation at 65,000 g for 45 min and the solubilized GPR120-G_i or GPR120-$G\alpha_{iq}$ complex was incubated with M1 anti-Flag affinity resin for 3 hours at 4°C. The resin was collected and washed with 20 column volumes of 20 mM HEPES pH 7.4, 100 mM NaCl, 2 mM $MgCl_2$, 5 mM $CaCl_2$, 10 μM TUG891 or 10 μM 9-HSA or 50 μM LA or 50 μM OA or 50 μM EPA, 0.01% (w/v) LMNG and 0.001% (w/v) CHS. The GPR120-G_i or GPR120-$G\alpha_{iq}$ complex was then eluted with buffer containing 20 mM HEPES pH 7.4, 100 mM NaCl, 2 mM $MgCl_2$, 10 μM TUG891 or 10 μM 9-HSA or 50 μM LA or 50 μM OA or 50 μM EPA, 0.01% (w/v) LMNG, 0.001% (w/v) CHS, 5 mM EGTA and 0.2 mg/ml Flag peptide. Finally, the complex was concentrated and loaded onto a size-exclusion chromatography on a Superose 6 Increase 10/300GL column (GE Healthcare) equilibrated with the buffer containing 20 mM HEPES, pH 7.5, 50 mM NaCl, 3 mM $MgCl_2$, 0.0005% LMNG, 0.00001% CHS and 10 μM TUG891or 10 μM 9-HSA or 50 μM LA or 50 μM OA or 50 μM EPA. The peak fractions of GPR120-G_i or GPR120-$G\alpha_{iq}$ complexes were collected and concentrated to 8 mg/ml using a 100 kDa cut-off Amicon Ultra Centrifugal Filter (Millipore) for making cryo grid.

Cryo-EM grid preparation and data collection

The purified GPR120-G_i or GPR120-$G\alpha_{iq}$ complex samples (concentrated to approximately 8 mg/ml) were applied onto holey carbon grids (Quantifoil, R1.2/1.3, 300 mesh), which were glow-discharged at 25 mA for 1 min using PELCO easiGlow. Excess samples were blotted for 3.5 s with a blot force of 8 with Ted Pella filter papers (catalog number: 47000-100). Afterward, the grids were vitrified by plunging into liquid ethane using a Vitrobot Mark IV (Thermo Fischer Scientific). Cryo-EM movies of all the six samples was performed on a Titan Krios at 300 kV in the Center of Cryo-Electron Microscopy, Zhejiang University (Hangzhou, China). Micrographs were recorded using a Gatan K2 Summit detector in counting mode with a pixel size of 1.014 Å using the SerialEM software (*42*). Movies were obtained at an exposure dose rate of about 8.0 e/Å^2/s with a defocus ranging from −0.5 to −2.5 μm. A total of 2,600, 1,992, 6,075, 5,441 and 4,844 movies were collected for the TUG891-, 9-HSA-, LA-, EPA- and OA-bound GPR120-G_i complexes, respectively.

For TUG891-bound GPR120-$G\alpha_{iq}$ complex, cryo-EM data collection was performed on a Titan Krios at 300 kV accelerating voltage at the Core Facilities, Zhejiang University Medical Center/Liangzhu laboratory. Micrographs were recorded using a Falcon 4 direct electron detector at a pixel size of 0.93 Å with EPU software. Image stacks were obtained at an exposure dose rate of about 8.7 electrons per Å^2 per second with a defocus ranging from −1.0 to −2.0 μm. The total exposure time was 6 s, resulting in a total exposure of 52 electrons per Å^2. A total of 7,093 movies were collected for TUG891-bound GPR120-$G\alpha_{iq}$ complex.

Cryo-EM data processing

Cryo-EM image stacks were aligned using MotionCor2.1 (*43*). Contrast transfer function (CTF) parameters were estimated by Gctf-v1.18 (*44*). The following data processing was performed using Relion-3.0.8 (*45*), Relion-4.0-beta (*45*, *46*) and CryoSPARC4.0.3 (*47*). For the TUG891-, 9-HSA-, LA-, and OA-bound GPR120-G_i complexes, automated particle picking using Gaussian blob detection yielded particles that were subjected to reference-free 2D classifications (classes=100; Regularisation parameter, T=2; mask diameter=160 Å) or 3D classifications (classes=4 or 6; T=4; mask diameter=160 Å) to discard bad particles for further processing. The map of neurotensin receptor 1-G_{i1} complex (EMD-20180) or the density map of TUG891-bound complex low-pass filtered to 40 Å were used as the reference maps for 3D classifications. Further focused 3D classifications (classes=3 to 6; T=4; mask diameter=160 Å) produced well-defined subsets which were subsequently subjected to 3D refinement (initial low-pass filter=20 Å; mask diameter=160 Å), CTF refinement and Bayesian polishing. The final refinement generated maps with indicated global resolutions of 3.0 Å (TUG891-bound complex), 3.1 Å (9-HSA-bound complex), 3.1 Å (LA-bound complex) and 2.8 Å (OA-bound complex) at a Fourier shell correlation of 0.143. Local resolution was determined using the Bsoft package with half maps as input maps. For OA-bound GPR120-Gi complex, the final particles were subjected to another round of local refinement with a soft mask on the 7TM region of GPR120 to further improve the map quality of receptor region.

For EPA-bound GPR120-G_i complex and TUG891-bound GPR120-G_{iq} complex, the picked particle projections were imported to CryoSPARC for 2D classification (classes=100; batchsize =200) to discard poorly defined particles. The selected particle projections were further subjected to ab-initio reconstruction and heterogeneous refinement (classes=3 or 4; batchsize =2000) in CryoSPARC. The well-defined subsets were re-extracted for further processing in Relion, which were subsequently subjected to 3D refinement (initial low-pass filter=20 Å; mask diameter=160 Å), CTF refinement and Bayesian polishing. The following non-uniform refinement of the complexes in CryoSPARC generated a map with an indicated global resolution of 2.81 Å (EPA-bound) and 2.64 Å (TUG891-bound) at a Fourier shell correlation of 0.143. To further improve the map quality of the complex especially for receptor, local 3D reconstruction (batchsize=200) focusing on the receptor and G protein complex were performed in CryoSPARC. The locally refinement maps for the receptor and G protein complex were combined based on the global refinement map using “vop maximum” command in UCSF Chimera.

Model building and structure refinement

The initial homology model of GPR120 receptor was generated using SWISS-MODEL (*48*). The initial coordinates of G_{i1} and scFv16 were generated from the Dopamine D3 receptor-G_{i1}-scFv16 complex (PDB ID: 7CMV). Agonists coordinates and geometry restraints were generated using phenix.elbow (*49*). Models were docked into the EM density map using UCSF Chimera (*50*) and Chimera X (*51*, *52*). This initial model was then subjected to iterative rounds of manual adjustment and automated refinement in Coot (*53*) and real-space refinement in Phenix (*49*), using minimization_global, local_grid_search, and ADP refinement with secondary structure and Ramachandran restraints. The final refinement statistics were validated using the module “comprehensive validation (cryo-EM)” in Phenix (*49*). The final refinement statistics are provided in table S5. The EM densities for these structures enabled the construction of models of major receptor and G protein regions (see table S25 for details). Helix 8 is not visible in the GPR120 receptor due to poor EM density. For those residues with ambiguous side-chain densities, corresponding side-chain atoms were removed from the final model of EPA-GPR120-G_i, 9-HSA-GPR120-G_i, LA-GPR120-G_i, OA-GPR120-G_i, and TUG891-GPR120-G_{iq} complexes; All those residues were summarized and listed in table S26. To avoid overstatement of the potential interactions, all those indicated residues were not described or discussed in our manuscript.

Gα-Gγ dissociation assay

The gene of wild-type human GPR120 short splicing isoform (hGPR120-S) or GPR120 long splicing isoform (hGPR120-L) was cloned into pcDNA3.1 vector with N-terminal HA signal sequence and FLAG tag for Gα-Gγ dissociation assays in HEK293 cells. G protein bioluminescence resonance energy transfer (BRET) probes, including Gα-Nluc [$G\alpha_i$- Nluc (insertion of Nluc after Ala87 in $G\alpha_i$), $G\alpha_q$- Nluc (insertion of Nluc after Leu97 in $G\alpha_q$) and $G\alpha_s$- Nluc (insertion of Nluc after Leu113 in $G\alpha_s$)], Gβ1, and Gγ2-GFP10, were generated according to previously described (*36*, *54*, *55*). HEK293 cells were transiently co-transfected with GPR120 WT or mutants and G protein BRET probes. Twenty-four hours after transfection, cells were distributed into 96-well microplates at a density of 5×10^4 cells per well and incubated for another 24 hours at 37°C. The cells were washed twice with Tyrode’s buffer (140 mM NaCl, 2.7 mM KCl, 1mM $CaCl_2$, 12 mM $NaHCO_3$, 5.6 mM D-glucose, 0.5 mM $MgCl_2$, 0.37 mM NaH_2PO_4 and 25 mM HEPES, pH7.4) and stimulated with different FAs at different concentrations for 2 min. BRET signal was measured after the addition of the luciferase substrate coelenterazine 400a (5 μM) using a Mithras LB940 microplate reader equipped with BRET filter sets. The BRET signal was calculated as the ratio of light emission at 510 nm/400 nm.

β-arrestin1/β-arrestin2 recruitment assay

The gene of wild-type human GPR120 short splicing isoform (hGPR120-S) or GPR120 long splicing isoform (hGPR120-L) was cloned into pcDNA3.1 vector with N-terminal HA signal sequence and FLAG tag and a C-terminal YFP tag for β-arrestin1/β-arrestin2 recruitment assays in HEK293 cells. HEK293 cells were transiently co-transfected with β-arrestin1-Rluc or β-arrestin2-Rluc and YFP-tagged receptor (GPR120 WT or mutants) as previously described (*23*, *54*, *56*). Twenty-four hours after transfection, the cells were distributed into 96-

well microplates at a density of 5×10^4 cells per well. After another twenty-four hours incubation at 37°C, the cells were washed twice with Tyrode's buffer and stimulated with different FAs at different concentrations for 10 min. Luciferase substrate coelenterazine-h (5 μM) was added before light emissions were recorded using a Mithras LB940 microplate reader (Berthold Technologies) with BRET filter sets. The BRET signal was determined by calculating the ratio of luminescence at 530 nm/485 nm.

cAMP accumulation assay

To examine the intracellular cAMP levels of HEK293 cells overexpressing GPR120-S or GPR120-L in response to different ligands, the GloSensor-based cAMP accumulation assay was performed as described previously (*23*, *57*, *58*). The similar cell surface expression levels of GPR120 WT or mutants were accomplished by adjusting the amount of plasmid used for transfection, and confirmed by ELISA and Western blot. In brief, pcDNA3.1-GPR120-S, pcDNA3.1-GPR120-L, or control vehicle pcDNA3.1 was co-transfected with the GloSensor plasmid into HEK293 cells with PEI, followed by incubation in six-well plates at 37°C for 24 hours, 5% CO_2 atmosphere. Twenty-four hours after transfection of GPR120 WT or corresponding mutants, the cells were seeded into 96-well microplates at a density of 5×10^4 cells per well. After another twenty-four hours incubation at 37°C, the cells were washed twice with Tyrode's buffer. Fifty microliters of serum-free DMEM containing fluorogenic substrate was added to each well of the 96-well plate 2 hours before measurement and stimulated with different FAs. The cells were pre-treated with 100 ng/ml PTX for 16 hours before stimulation with different FAs. Luminescence signal was measured using an EnVision multimode plate reader (Perkin Elmer).

Inositol triphosphate measurement

The intracellular inositol triphosphate (IP3) levels of HEK293 cells overexpressing GPR120-S or GPR120-L in response to TUG891 or other GPR120 agonists were evaluated using IP3 kit (Cloud-Clone, CEC037Ge) according to the manufacturer's protocol. Briefly, HEK293 cells were transfected with either GPR120-L, GPR120-S or control vehicle pcDNA3.1 and incubated at 37°C, 5% CO_2 atmosphere. Forty-eight hours post transfection, the cells were washed twice with Tyrode's buffer and stimulated with TUG891 at different concentrations for 20 min. The cells were then lysed by the lysis reagent provided in the ELISA Kit (Cloud-Clone) and the intracellular IP3 levels were measured according to the manufacturer's instructions. Luminescence intensity at 450 nm was measured using an EnVision multimode plate reader (Perkin Elmer).

Data analysis for bias properties

For direct comparison of GPR120 ligands on particular signaling (effector) pair, the "Operational Model" was used as previously described (*24*, *59*). The coupling efficiency (τ) of selected GPR120 ligand to designated downstream signaling pathways is calculated by fitting the concentration-dependent data into the Eq. 1 using GraphPad Prism 8 as previously described (*24*)

$$\frac{E}{E_{\mathrm{m}}}=\frac{\tau[A]}{\tau[A]+([A]+K_{\mathrm{D}})} \tag{1}$$

where $[A]$ and E are the concentration and efficacy of the designated ligand, K_D is the dissociation constant of the ligand determined from a FITC-ligand competition binding assay; E_m is the maximal efficacy response (E_{max}) by a full agonist in this system.

The effective signaling (σ) for ligand (lig) over a selected reference compound (ref) in particular signaling pathway can be calculated using Eq. 2 as previously described (*24*).

$$\sigma_{\mathrm{lig}}=\log\left(\frac{\tau_{\mathrm{lig}}}{\tau_{\mathrm{ref}}}\right) \tag{2}$$

FITC-ligand competition binding assay

To determine the dissociation constant of individual ligands to GPR120, a bioluminescence resonance energy transfer (BRET)-based ligand competition binding assay was used as described previously (*23*, *57*). In brief, the plasmid containing GPR120 with a Nluc fused at its N terminus (pcDNA3.1-Nluc-GPR120) was transiently transfected into HEK293 cells with Lipofectamine 2000 (Thermo Fisher). Twenty-four hours post transfection, cells were seeded into a black 96-well plate (3.5×10^4 cells/well) and further incubated for 24 hours. The saturation binding curve of FITC-TUG891 (FITC conjugated labeled at the carboxyl end of the TUG891) to GP120 was determined by treating the transfected cells with FITC-TUG891 ranging from 0 to 20 μM and with or without the presence of 200 μM TUG891 for 40 min at 37°C (see fig. S2A). For the competition experiments, the transfected cells were incubated with 2.5 μM of FITC-TUG891 and various concentrations of target competing ligands in HBSS buffer for 40 min at 37°C (see fig. S2, B to F). Luciferase substrate coelenterazine-h (5 μM) was added before light emissions were recorded using a Mithras LB940 microplate reader (Berthold Technologies) with BRET filter sets. The BRET signal was determined by calculating the ratio of luminescence at 535 nm/460 nm.

The inhibition constants (K_i) for respective ligands were calculated using Eq. 3 as previously described (*23*)

$$K_{\mathrm{i}}=\frac{\mathrm{IC}_{50}}{1+\frac{[\mathrm{L}]}{K_{\mathrm{d}}}} \tag{3}$$

where [L] is the FITC-TUG891 concentration (nM), and the K_d is the dissociation constant of FITC-TUG891 generated from saturation binding experiment.

Enzyme-linked immunosorbent assay

To evaluate cell surface expression of the wild-type GPR120 or its mutants, HEK293 cells were transfected the plasmid of GPR120 wild-type or mutants using Lipofectamine 2000 (Thermo Fisher Scientific) as previously described (*23*, *36*). Twenty-four hours post transfection, cells were divided into 24-well plates at a density of 2×10^5 cells per well and further incubated for 24 hours at 37°C in 5% CO_2. The cells were washed three times with PBS, then fixed by 4% (w/v) paraformaldehyde for 5 min at room temperature, blocked with 5% (w/v) BSA for another 1 hour. Afterwards, the cells were washed and incubated with the mouse monoclonal anti-FLAG M2 primary antibody (Sigma-Aldrich, Cat# F1804, 1:1000) at 4°C for 1 hour, and incubated with secondary goat anti-mouse antibody (Thermo Fisher, Cat# A-21235, 1:5000) for 1 hour at room temperature. After washing, TMB (3,3′,5,5′-tetramethylbenzidine) solution was added for color reaction, which was then stopped by adding an equal volume of 0.25 M HCl. The absorbance at 450 nm was determined using a TECAN luminescence counter (Infinite M200 Pro NanoQuant). Similar cell surface expression levels of corresponding GPR120 or its mutants determined by ELISA assay.

Western blotting analysis

Comparative plasma membrane expression levels of GPR120-S and GPR120-L, or GPR120 mutants were accomplished by adjusting of the amount of plasmids transfected into in HEK293 cells, and confirmed by Western blotting. HEK293 cells were transiently transfected with pcDNA3.1-GPR120-S, pcDNA3.1-GPR120-L, pcDNA3.1-GPR120-S-YFP, pcDNA3.1-GPR120-L-YFP, or control vehicle plasmid pcDNA3.1, respectively. Forty-eight hours post transfection, the transfected HEK293 cells were washed in pre-colded HBSS buffer twice. The cells were then scraped and pelleted by centrifugation at 600 g at 4°C for 5 min. The cell pellets were disrupted by homogenization, and centrifugated at 600 g at 4°C for 30 min to remove fractions of nuclei and cell apparatus. To collect cell membranes, the supernatant was further centrifugated at 60,000 g at 4°C for 30 min. The collected cell membranes were homogenized and solubilized with HBSS buffer containing 1% (w/v) DDM and protease inhibitor cocktail (Roche, 11697498001) for 2 h at 4 °C with end-to-end rotation. After removing the insolubilized precipitate with centrifugation at 60,000 g for 30 min, the supernatant was subjected to SDS-PAGE and Western blot for further analysis. For detection of the GPR120 and ATP1A1, the anti-FLAG (Invitrogen, PA1-984B) or anti-ATP1A1 (proteintech, 14418-1-AP) primary antibodies were used at 1:1,000 dilution, respectively. The HRP-conjugated goat anti-rabbit IgG (Sigma Aldrich, A6154) was used as the secondary antibody at a 1:5,000 dilution.

Molecular dynamics simulations

To verify the stability of the various ligands in the binding pocket of GPR120, we performed molecular dynamics (MD) simulations of the OA-GPR120-G_i, LA-GPR120-G_i, EPA-GPR120-G_i, 9-HSA-GPR120-G_i, TUG891-GPR120-G_i, and TUG891-GPR120-G_{iq} wild system. The CHARMM36m force field (*60*) was used for OA-GPR120-G_i, LA-GPR120-G_i, EPA-GPR120-G_i, TUG891-GPR120-G_i, 9-HSA-GPR120-G_i, TUG891-GPR120-G_{iq}, lipids, ions and the TIP3P model water molecules, as described by our previous studies (*36*). All of the MD input files were generated by CHARMM-GUI (*61*) website, in which the force field parameter of ligands were generated by CHARMM General Force Field (CGenFF) (*62*). The structures of the OA-GPR120-G_i, LA-GPR120-G_i, EPA-GPR120-G_i, 9-HSA-GPR120-G_i, TUG891-GPR120-G_i, and TUG891-GPR120-G_{iq} complexes were embedded into a pre-equilibrated and periodic structure of 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphatidylcholine membrane. Then, 0.15 M NaCl was added to balance the charge of the OA-GPR120-G_i, LA-GPR120-G_i, EPA-GPR120-G_i, TUG891-GPR120-G_i, 9-HSA-GPR120-G_i, and TUG891-GPR120-G_{iq} complex system using the Monte Carlo Method (*63*). Next, the systems were solvated into periodic hexagonal type TIP3P water box with a size of around 80 Å × 80 Å × 100 Å by replacement methods using the membrane orientation calculated by the Orientations of Proteins in Membranes database. The OA-GPR120-G_i, LA-GPR120-G_i, EPA-GPR120-G_i, 9-HSA-GPR120-G_i, TUG891-GPR120-G_i, and TUG891-GPR120-G_{iq} complex systems were first processed for energy minimization for 10,000 steps using gmx mdrun module in Gromacs2019.6 (*64*) software, of which the first 5,000 steps were performed using the steepest descent method and the remaining 5,000 steps using the conjugated gradient method. The OA-GPR120-G_i, LA-GPR120-G_i, EPA-GPR120-G_i, 9-HSA-GPR120-G_i, TUG891-GPR120-G_i, and TUG891-GPR120-G_{iq} complex systems were heated from 0 K to 310 K in the NVT ensemble (number of particles, volume and temperature are conserved) for over 1,000 ps. Next, production simulations were run at 1 atm in the NPT ensemble (number of particles, pressure and temperature are conserved) for over 1,000 ps with 10.0 kcal mol^{-1} Å^{-2} harmonic restraints. The NVT ensemble, NPT ensemble and the 500 ns MD production with a time step of 2 fs were performed using Gromacs2019.6. The particle mesh Ewald method was used to calculate electrostatic interactions with a cut-off of 12 Å. The bonds involving hydrogen were kept fixed using the SHAKE algorithm during each integration time step of 2 fs. All analyses were completed in Gromacs2019.6 and VMD (*65*).

To investigate the effect of π:π interactions between double bonds and aromatic residues on ligand recognition by GPR120, we constructed eight systems that the double bond vi in LA-GPR120 was mutated to a single bond vi, the double bond ix in OA-GPR120 was mutated to a single bond ix, the double bond iii or double bond xii or double bond xv in EPA-GPR120 was mutated into single bond iii or single bond ix or single bond xv. We also constructed additional sixteen systems that the F28 or F88 or F177 or W277 in LA-GPR120 was mutated to L28 or L88 or L177 or L277, the F28 or F115 or F303 in OA-GPR120 was mutated to L28 or L115 or L303, the F27 or F28 or F115 or F177 or W198 or W207 or F211 W277 or F303 in EPA-GPR120 was mutated to L27 or L28 or L115 or L177 or L198 or L207 or L211 or W277 or L303.

To further investigate the receptor-G protein interactions according to our experimental TUG891-GPR120-G_{iq} complex, we performed MD simulations for TUG891-GPR120-G_q complex using our experimental model as the initial template. In our MD simulations, we mutated the residues on the loop connecting helix 4 (H4) and β-sheet 6 (S6) of $G\alpha_{i1}$, including $K312^{G.h4s6.03}$, $R313^{G.h4s6.04}$, $K314^{G.h4s6.08}$, $D315^{G.h4s6.09}$, $T316^{G.h4s6.10}$, to Pro, Asp, Ser, Asp of the corresponding $G\alpha_q$ sequence. We also mutated $E308^{G.h4.16}$ and $E318^{G.h4s6.12}$ of $G\alpha_{i1}$ to their corresponding G_q residues Val and Ile, respectively. The interactions observed by experiments and MD simulations were further validated by extensive mutagenesis analysis. For analysis of the coupling of G_s to GPR120, the receptor region of EPA-GPR120-G_i complex and G_s region of the ADGRG2-β-G_s complex (PDB ID: 7WUI) were extracted and merged to construct the EPA-GPR120-G_s complex. The same molecular dynamics simulation method was used for the 9-HSA-GPR120-G_i, OA-GPR120-G_i, LA-GPR120-G_i, EPA-GPR120-G_i, TUG891-GPR120-G_i, and TUG891-GPR120-G_{iq} complexes.

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SUPPLEMENTARY MATERIALS
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Figs. S1 to S32
Tables S1 to S26
MDAR Reproducibility Checklist

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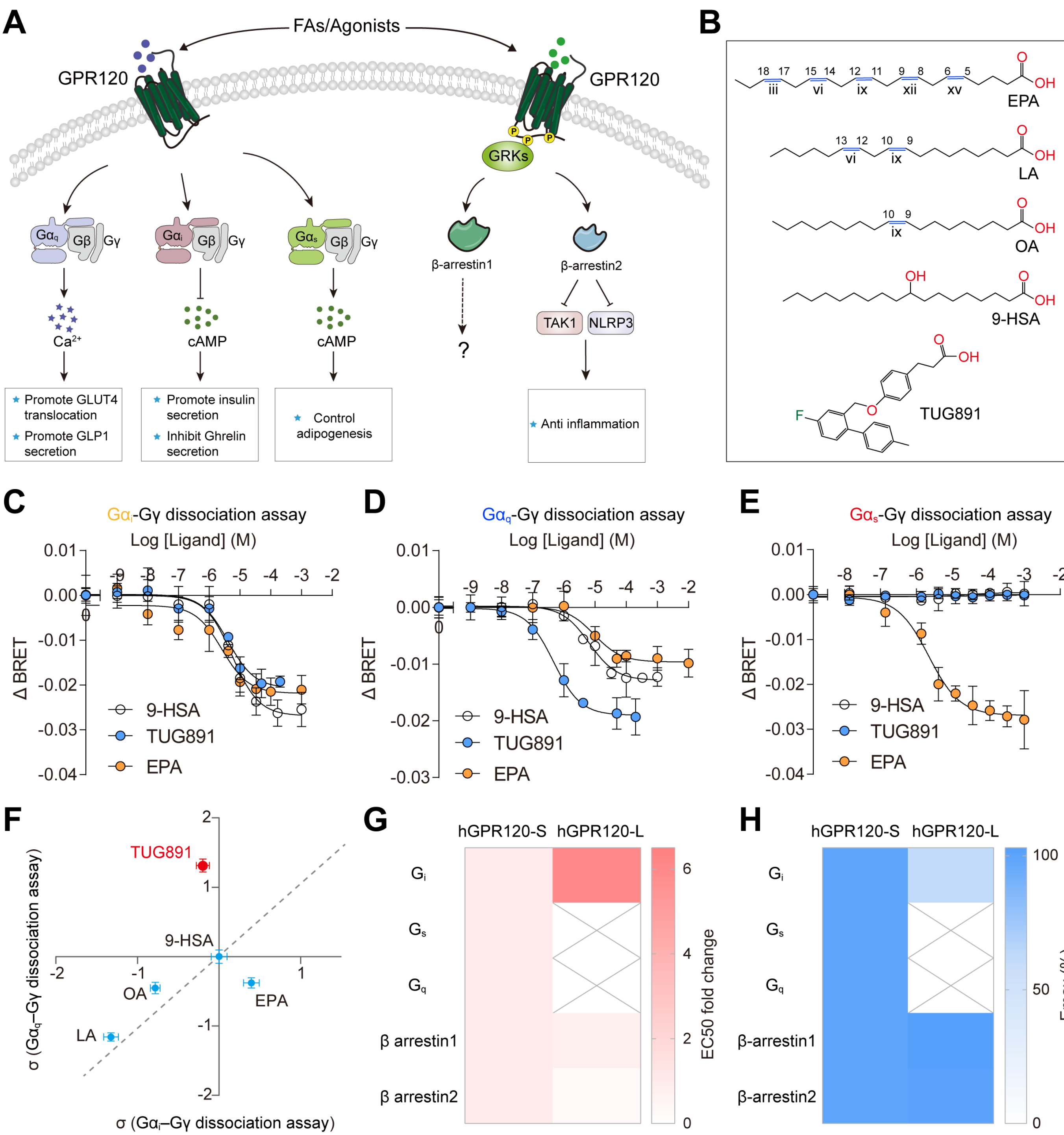
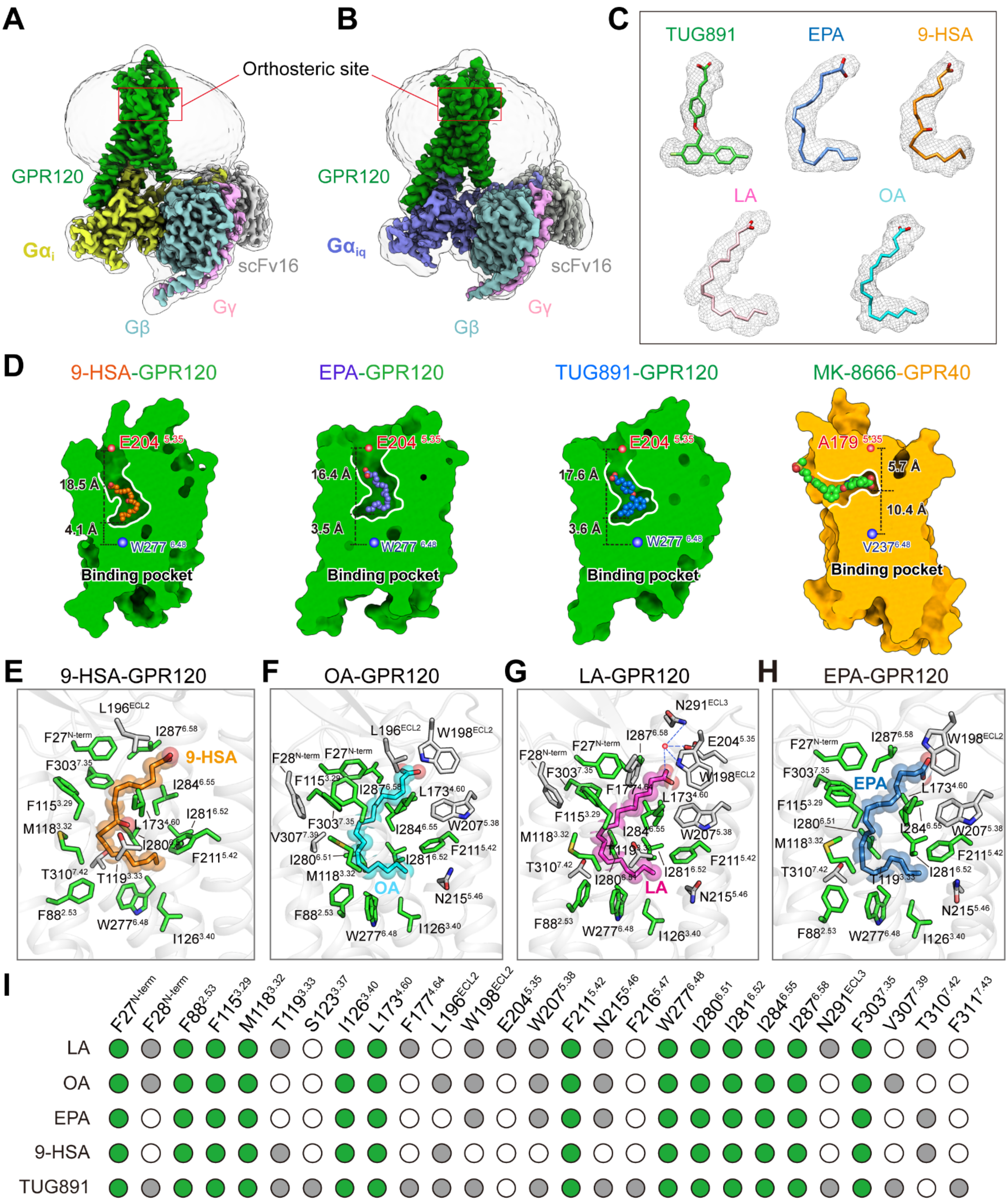
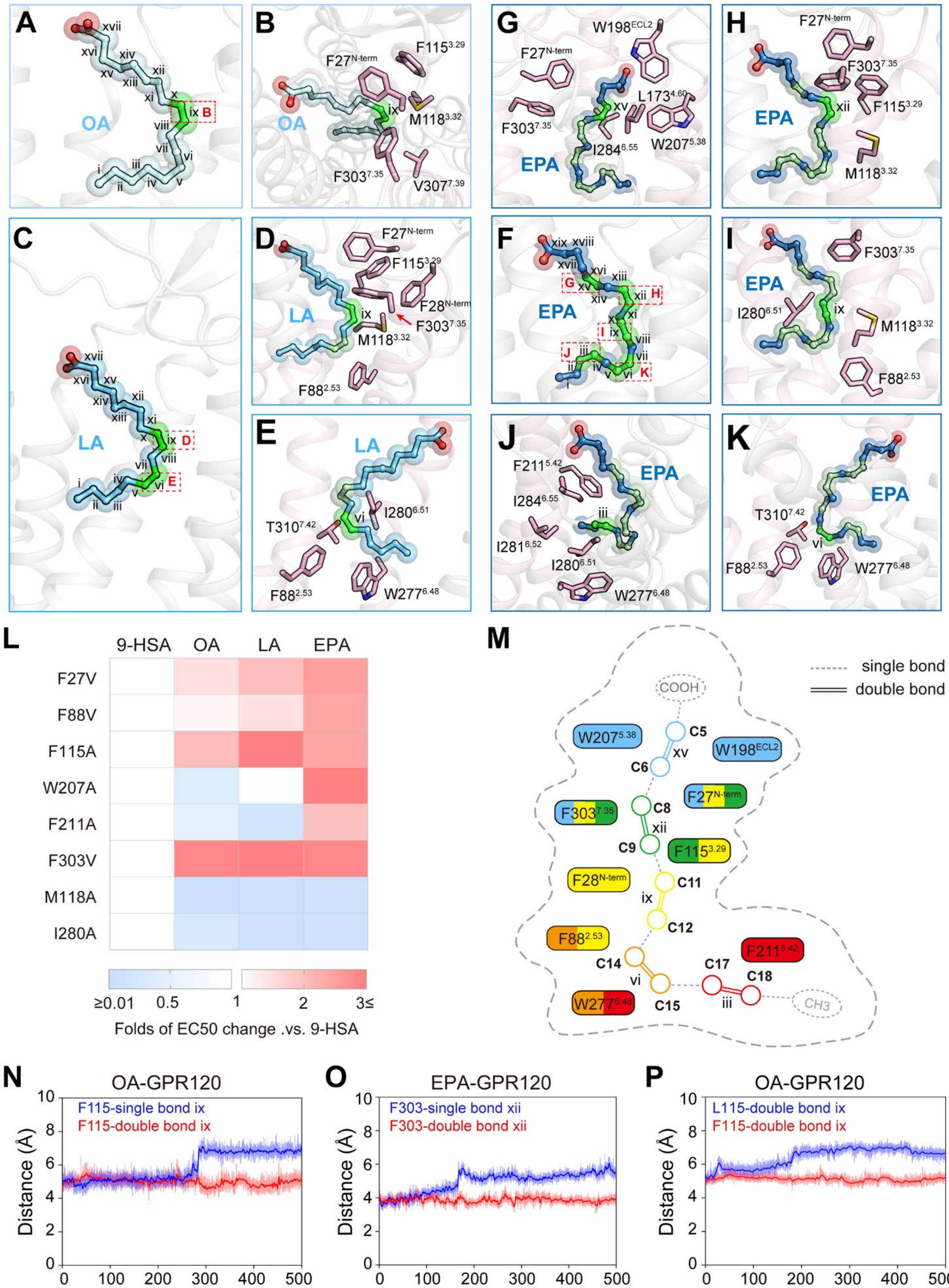

A
FAs/Agonists
GPR120
GPR120
GRKs
β-arrestin1
β-arrestin2
TAK1
NLRP3
?
cAMP
cAMP
Promote GLUT4 translocation
Promote GLP1 secretion
Promote insulin secretion
Inhibit Ghrelin secretion
Control adipogenesis
Anti inflammation
B
EPA
LA
OA
9-HSA
TUG891
C
Gαi-Gγ dissociation assay
D
Gαq-Gγ dissociation assay
E
Gαs-Gγ dissociation assay
Log [Ligand] (M)
Δ BRET
9-HSA
TUG891
EPA
F
σ (Gαq–Gγ dissociation assay)
σ (Gαi–Gγ dissociation assay)
G
hGPR120-S
hGPR120-L
β arrestin1
β arrestin2
EC50 fold change
H
β-arrestin1
β-arrestin2
Emax (%)

Fig. 1. Biased properties of different GPR120 ligands. (**A**) Schematic overview of G protein and arrestin mediated GPR120 signaling and related functions. (**B**) Chemical structures of fatty acid ligands and synthetic ligand TUG891. (**C** to **E**) Concentration dependent response curves of GPR120 in response to ligands by (C) $G\alpha_i$-$G\gamma$ dissociation assay (D), $G\alpha_q$-$G\gamma$ dissociation assay (E), or $G\alpha_s$-$G\gamma$ dissociation assay. Values are mean ± standard deviation (SD) from three independent experiments (n=3) performed in triplicates. (**F**) Two-dimensional differential σ plot comparing effective signaling (σ) in $G\alpha_q$-$G\gamma$ dissociation and $G\alpha_i$-$G\gamma$ dissociation for a set of GPR120 ligands (9-HSA, TUG891, OA, LA, and EPA) using saturated FA 9-HSA as the reference. Ligands on the gray dashed diagonal line showed balanced signaling properties. The plot was generated based on σ values shown in table S2. (**G** and **H**) Heatmap representation of activities comparison between two splice variants of GPR120 in response to EPA stimulation, via G protein dissociation assays or β-arrestin1/2 recruitment assay. Heatmap shows the EC50 (G) and Emax (H) differences. The heatmaps are generated according to the data shown in fig. S1 (refer to table S4 for details). The hGPR120-L showed no detectable G_s or G_q activity and had weaker G_i coupling efficacy compared with hGPR120-S.



A
B
C
Orthosteric site
GPR120
Gαi
Gβ
Gγ
scFv16
GPR120
Gαiq
Gβ
Gγ
scFv16
TUG891
EPA
9-HSA
LA
OA
D
9-HSA-GPR120
EPA-GPR120
TUG891-GPR120
MK-8666-GPR40
E204 5.35
18.5 Å
4.1 Å
W277 6.48
Binding pocket
E204 5.35
16.4 Å
3.5 Å
W277 6.48
Binding pocket
E204 5.35
17.6 Å
3.6 Å
W277 6.48
Binding pocket
A179 5.35
5.7 Å
10.4 Å
V237 6.48
Binding pocket
E
9-HSA-GPR120
F
OA-GPR120
G
LA-GPR120
H
EPA-GPR120
I
LA
OA
EPA
9-HSA
TUG891

Fig. 2. Overall cryo-EM structures of the GPR120-G_i/G_{iq} heterotrimer bound with LA, OA, EPA, 9-HSA, TUG891 and the intramolecular interaction between ligands and its binding pocket. (**A** to **C**) Cryo-EM density map of - representative GPR120-G_i (exemplified by 9-HSA-GPR120-G_i complex) (A) and TUG891-GPR120-G_{iq} (B) complexes. Ligands with different colors from TUG891-GPR120-G_{iq}, EPA-GPR120-G_i, 9-HSA-GPR120-G_i, LA-GPR120-G_i, and OA-GPR120-G_i complexes (C) are shown, with surrounding density maps. TUG891, green; EPA, sky blue; 9-HSA, orange; LA, pink; OA, cyan. See also figs. S4H, S5F, S6F, S7G, S8G and S9G. (**D**) Binding of ligands 9-HSA (orange), LA (pink) and TUG891 (blue) in the ligand pockets of GPR120 from 9-HSA-GPR120-G_i, LA-GPR120-G_i and TUG891-GRP120-G_{iq} complexes (green), compared with binding of MK-8666 in GPR40 (PDB ID: 5TZR) (*25*) (orange). The depths of the ligand pockets were highlighted. The conserved "toggle switch" and the TM5 residue of E204$^{5.35}$ (GPR120)/A179$^{5.35}$ (GPR40) were used as reference. (**E** to **H**) The detailed interactions between 9-HSA (E), OA (F), LA (G), EPA (H) and residues in the ligand binding pocket of GPR120 in complex with G_i. The H-bonds and polar interactions are depicted as blue dashed line. Common interaction residues in all four ligands bound GPR120-G_i complex structures are shown in green, whereas specific interaction residues within corresponding models are shown in gray. Orange spheres, 9-HSA; Cyan spheres, OA; Magenta spheres, LA; Blue spheres, EPA; The red ball represents a water. (**I**) Barcode presentation of the binding pocket residues of GPR120 that contact with the ligands LA, OA, EPA, 9-HSA, and TUG891 in LA-, OA-, EPA-, 9-HSA-bound GPR120-G_i complexes and TUG891-bound GPR120-G_{iq} complex. The green dots show common interactions in five ligands. The gray dots show non-conserved interactions. The white dots mean there is no interaction between ligands and GPR120.



A
OA
B
F27N-term
F115 3.29
M118 3.32
F303 7.35
V307 7.39
C
LA
D
E
F
EPA
G
H
I
J
K
W198ECL2
L173 4.60
I284 6.55
W207 5.38
F28N-term
F88 2.53
I280 6.51
T310 7.42
W277 6.48
F211 5.42
I281 6.52
L
9-HSA
OA
LA
EPA
F27V
F88V
F115A
W207A
F211A
F303V
M118A
I280A
≥0.01
0.5
1
2
3≤
Folds of EC50 change .vs. 9-HSA
M
single bond
double bond
COOH
C5
C6
C8
C9
C11
C12
C14
C15
C17
C18
CH3
N
OA-GPR120
F115-single bond ix
F115-double bond ix
Distance (Å)
Time (ns)
O
EPA-GPR120
F303-single bond xii
F303-double bond xii
P
OA-GPR120
L115-double bond ix
F115-double bond ix

Fig. 3. Double bonds recognition by aromatic residues of GPR120. (**A** and **B**) GPR120 with bonds i-xvii (C1-C18) in OA (A) and the double bond ix (C9-C10) in OA (B). OA, pale cyan; double bond is highlighted in green. Bonds are labeled from i to xvii according to their position relative to the carbon end, and C atoms are labeled from C1 to C18 according to their position relative to the carboxyl terminal. Double bonds are highlighted in green. (**C** to **E**) GPR120 with bonds i-xvii (C1-C18) in LA (C) with residues around double bond ix (C9-C10) (D) or double bond vi (C12-C13) (E). LA, cyan. (**F** to **K**) GPR120 with bonds i-xix (C1-C20) in EPA (F) with residues around double bond xv (C5-C6) (G), double bond xii (C8-C9) (H); double bond ix (C11-C12) (I), double bond iii (C17-C18) (J), and double bond vi (C14-C15) (K). EPA, blue. (**L**) Differential effects of mutations of double bond contacting residues on unsaturated FAs (OA, LA, and EPA) compared with those on 9-HSA induced GPR120 activation by $G\alpha_i$-$G\gamma$ dissociation assay. Heatmap shows the folds of EC50 changes between designated ligand and 9-HSA. The heatmap was generated using data shown in fig. S17, A to G (refer to table S11 for details). (**M**) Schematic showing the recognition of double bonds at specific positions of FAs by aromatic amino acids of GPR120. The double bonds interacting aromatic residues of GPR120 are shown, and highlighted using color shaded boxes in corresponding to individual double bond. Here, the double bonds of EPA are shown as an example. Dashed single lines represent omitted carbon atoms and corresponding single bonds; double segments with different colors represent double bonds. Double bond iii, red; Double bond vi, orange; Double bond ix, yellow; Double bond xii, green; Double bond xv, blue. Refer to fig. S17H for details. The curved dotted line segment represents the ligand binding pocket. (**N** to **P**) The distance changes between F115 and bond ix in OA-GPR120 system (N), the distance change between F303 and bond xii in EPA-GPR120 system (O), the distance changes between L115 or F115 and bond ix in OA-GPR120 system (P) during a 500 ns molecular dynamics simulation. The red line represents the double bond ix system [(N) and (P)] double bond xii system (O) and the blue line represents the double bond ix (N), xii (O) was mutated to a single bond, or F115 was mutated to L115 system (P). The shaded area represents the original, unsmoothed value, with reduced transparency, and the thick line was the result of smoothing the original distance value every 200 points.

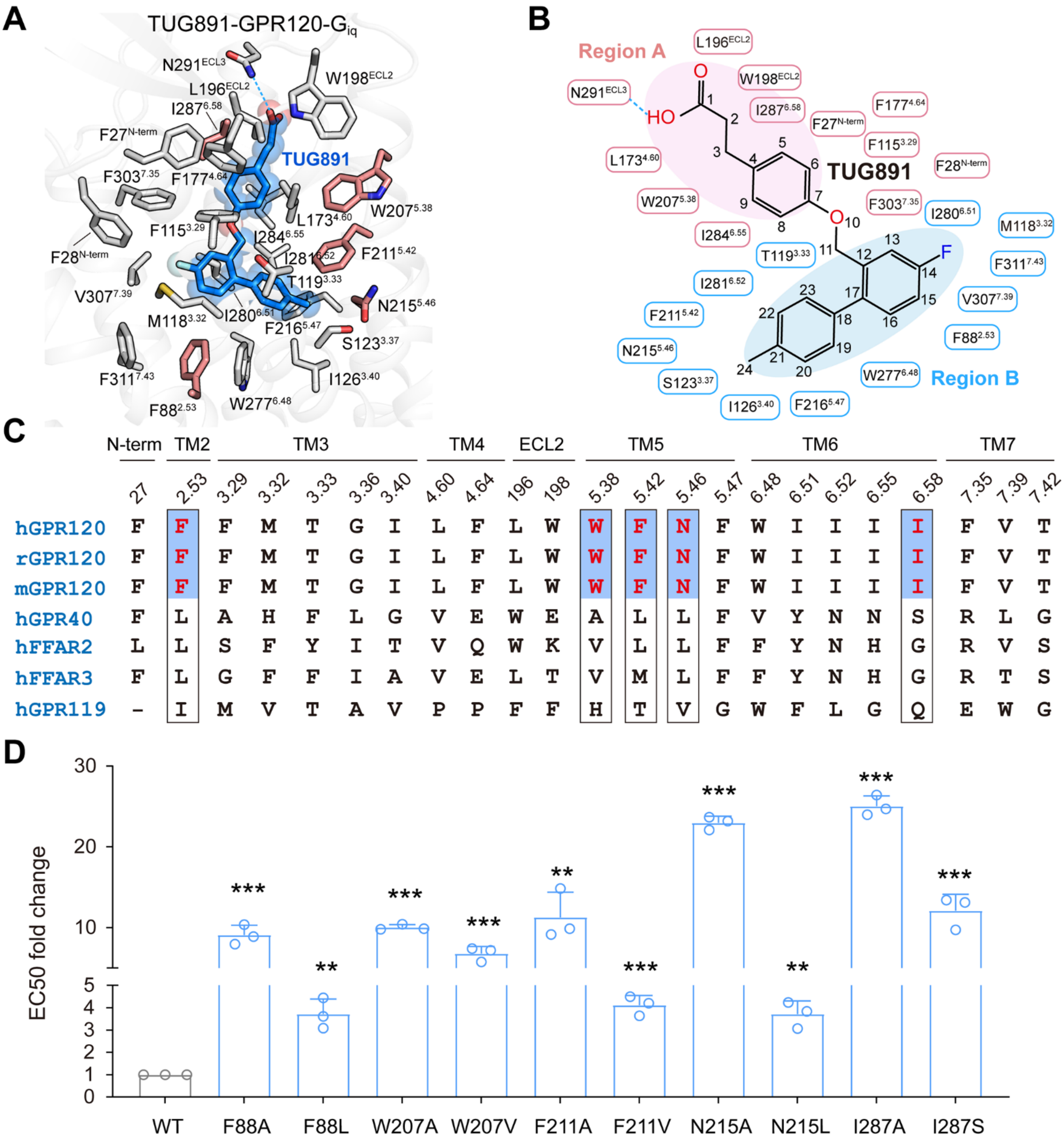

A
TUG891-GPR120-G_{iq}
TUG891
B
Region A
Region B
TUG891
C
N-term TM2 TM3 TM4 ECL2 TM5 TM6 TM7
27 2.53 3.29 3.32 3.33 3.36 3.40 4.60 4.64 196 198 5.38 5.42 5.46 5.47 6.48 6.51 6.52 6.55 6.58 7.35 7.39 7.42
hGPR120 F F F M T G I L F L W W F N F W I I I I F V T
rGPR120 F F F M T G I L F L W W F N F W I I I I F V T
mGPR120 F F F M T G I L F L W W F N F W I I I I F V T
hGPR40 F L A H F L G V E W E A L L F V Y N N S R L G
hFFAR2 L L S F Y I T V Q W K V L L F F Y N H G R V S
hFFAR3 F L G F F I A V E L T V M L F F Y N H G R T S
hGPR119 - I M V T A V P P F F H T V G W F L G Q E W G
D
EC50 fold change
WT F88A F88L W207A W207V F211A F211V N215A N215L I287A I287S

Fig. 4. Structural basis of ligand selectivity of TUG891. (**A**) Detailed interactions between TUG891 (blue) and residues in the binding pocket of GPR120 (gray and salmon) in TUG891-GPR120-G_{iq} complex structure. The residues highlighted in salmon are distinct compared to other GPCRs known to sense LCFAs as shown in Fig. 4C. H-bond depicted as a blue dashed line. (**B**) 2D representation of contacts between GPR120 and the ligand TUG891 shown in a. Residues that were implicated in interactions within the region A subpocket were presented in pink, within the region B subpocket were presented in blue. H-bond was depicted as a blue dashed line. (**C**) Sequence alignment of GPR120 from different species and with other lipid receptors. Note that the residues F88$^{2.53}$, W207$^{5.38}$, F211$^{5.42}$, N215$^{5.46}$ and I287$^{6.58}$ of TM2, TM5 and TM6 in GPR120 are unique compared to other lipid receptors. (**D**) Mutational effects of GPR120 pocket residues on TUG891 induced $G\alpha_i$-$G\gamma$ dissociation activities. Statistical differences between wild type (WT) and mutants were determined by two-sided, one-way analysis of variance (ANOVA) with Tukey's test. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; ns, no significant difference. Values are the mean ± SD of three independent experiments for the wild type (WT) and mutants (n=3).

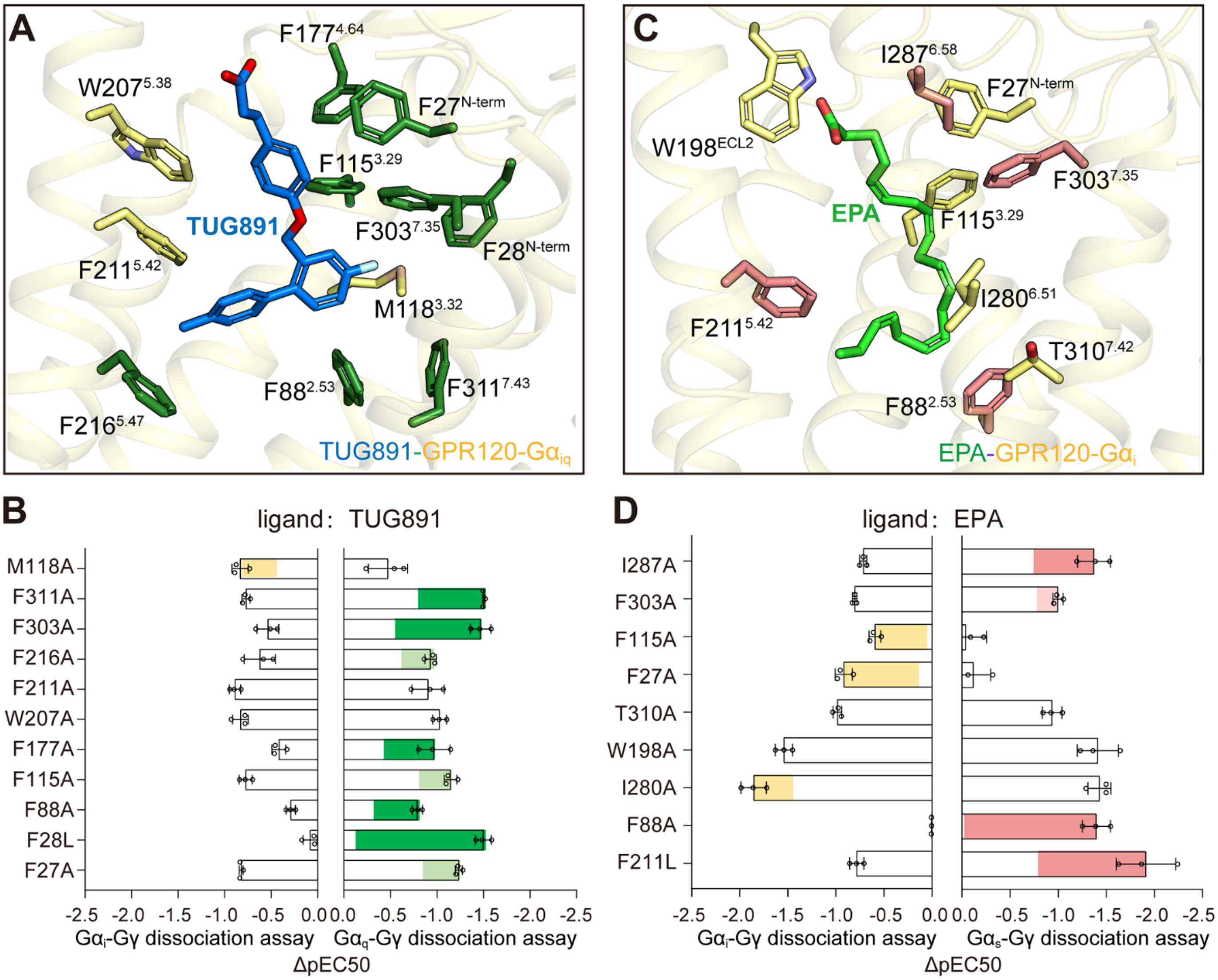


Fig. 5. Contribution of pocket residues to signaling bias of GPR120. (**A**) The detailed interactions between TUG891 and pocket residues in the ligand binding pocket of GPR120 in complex with G_{iq}. Residues showed larger effects on G_q activation than that of G_i are highlighted in green. TUG891, blue. (**B**) Mutational effects of the aromatic residues contacting with TUG891 or EPA double bonds on TUG891 induced GPR120-G_q or GPR120-G_i coupling activity measured by $G\alpha_q$-$G\gamma$ or $G\alpha_i$-$G\gamma$ dissociation assays, respectively. The bar graphs showed the differences of potency (ΔpEC50) between wild-type GPR120 and its mutants, which were generated according to data shown in fig. S22, B and C. Residues contributing to biased signaling to either G_i or G_q pathways were shown as light green (0 < ΔΔpEC50 < 0.5), green bars (0.5 < ΔΔpEC50) or yellow bars (ΔΔpEC50 < 0). ΔΔpEC50 is calculated by subtraction of the ΔpEC50 of G_q activity by ΔpEC50 of G_i activity. Values are mean ± SD from three independent experiments (n=3) performed in triplicates. (**C**) The detailed interactions between EPA and pocket residues in the ligand binding pocket of GPR120 in complex with G_i. Residues showed larger effects on G_s activation than that of G_i are highlighted in pink. EPA, green. (**D**) Mutational effects of residues contacting with EPA double bonds on EPA induced GPR120-G_s or GPR120-G_i coupling activity measured by $G\alpha_s$-$G\gamma$ or $G\alpha_i$-$G\gamma$ dissociation assays, respectively. The bar graphs showed the differences of potency (ΔpEC50) between wild-type GPR120 and its mutants, which were generated according to data shown in fig. S22, E and F. Residues contributing to biased signaling to either G_i or G_s pathways were shown as light pink (0 < ΔΔpEC50 < 0.5), pink bars (0.5 < ΔΔpEC50) or yellow bars (ΔΔpEC50 < 0). ΔΔpEC50 is calculated by subtraction of the ΔpEC50 of G_s activity by ΔpEC50 of G_i activity. Values are mean ± SD from three independent experiments (n=3) performed in triplicates.

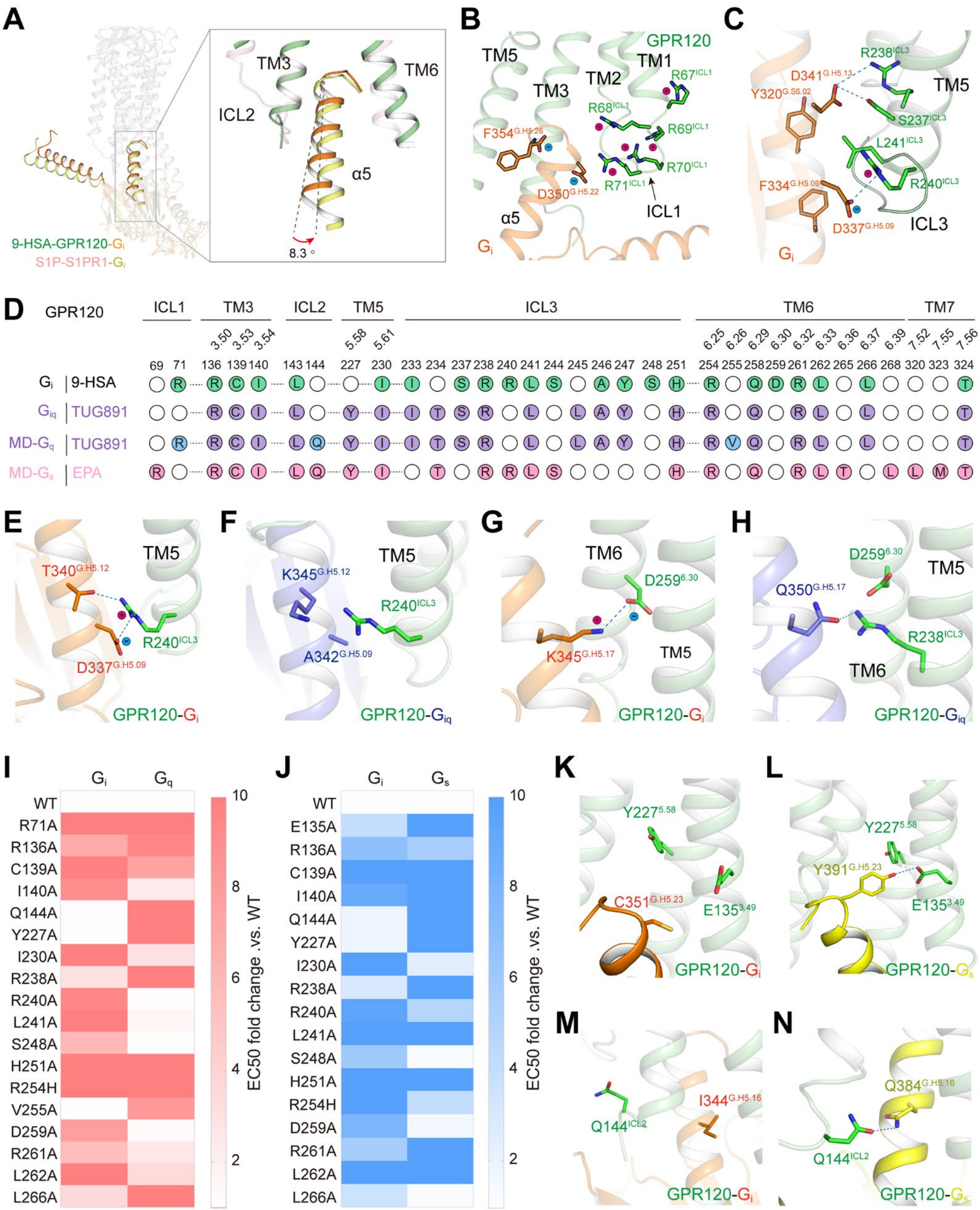
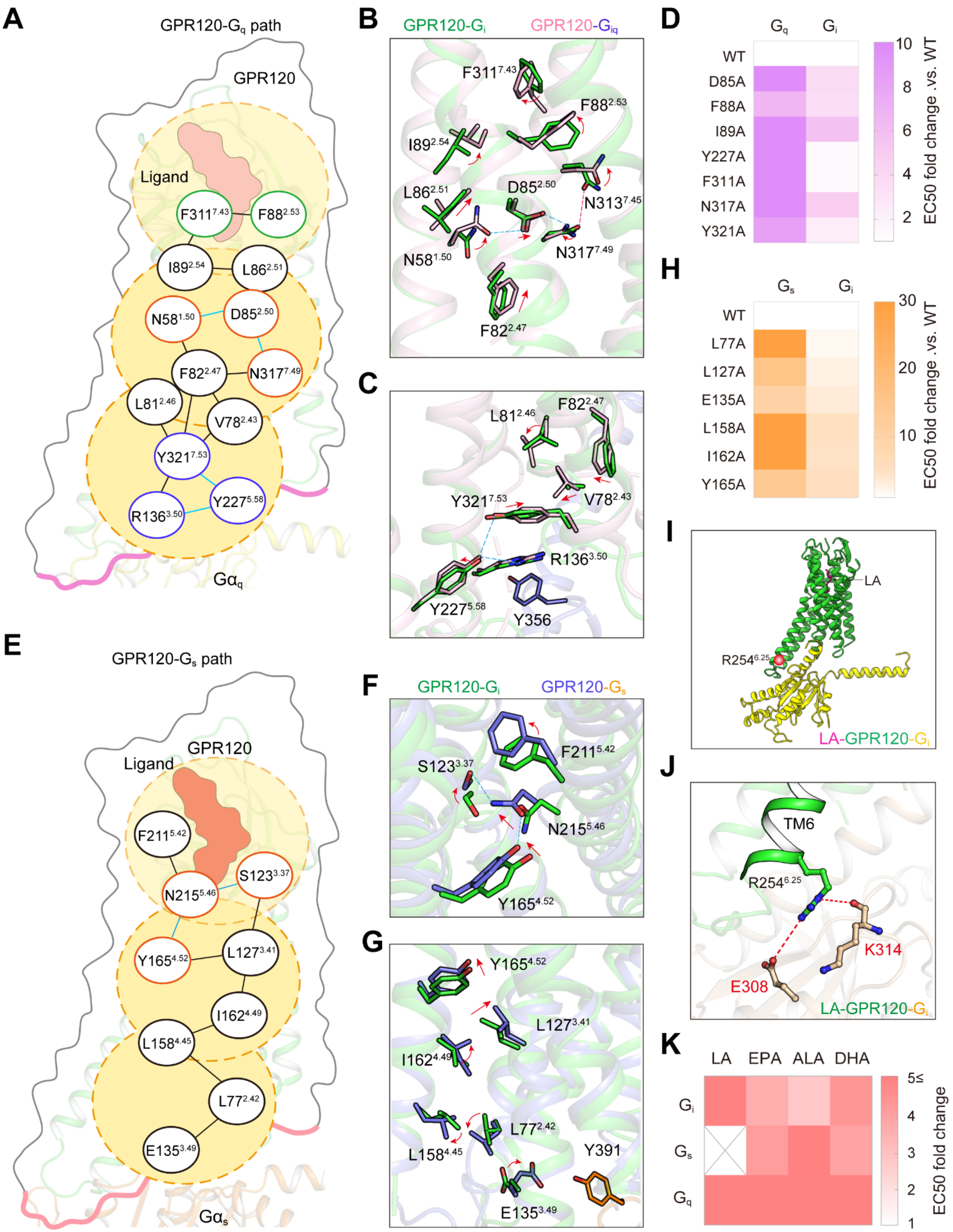


Fig. 6. GPR120-G_i interface and coupling of G_q or G_s to GPR120 by MD simulation. (**A**) Comparison of the overall interaction mode between 9-HSA-GPR120-G_i complex with that of the S1P-S1PR1-G_i complex structure (PDB: 7EVY) (*66*). The C-tail of the α5 helix was rotated away from the TM3 and toward TM6 by approximately 8°. (**B** and **C**) Interactions between residues on ICL1 and ICL2 of GPR120 and $G\alpha_i$. (B). The ICL3 residues of GPR120 form interactions with G_i (C). (**D**) Interaction differences in the interface of the 9-HSA-GPR120 with G_i, TUG891-GPR120 with G_q, and EPA-GPR120 with G_s in corresponding EM structures or simulated structures, respectively. Residues of GPR120 that interact with G_i were indicated by green circle. Residues of GPR120 that interact with G_q were indicated by purple circle. Residues of GPR120 that interact with G_s were indicated by pink circle. Residues of GPR120 showing no interaction with G_i/G_q/G_s were indicated by blank circle. Note that residues of R71, Q144 and V255 in GPR120 simulated structures formed interactions were verified by mutagenesis data and marked as blue circles. (**E** to **H**) Comparison of the TUG891-GPR120-G_{iq} and 9-HSA-GPR120-G_i complex structures revealed different interactions. (**I**) Heatmap representation of the effect of mutations in the GPR120 interface on G_i or G_q coupling in response to 9-HSA stimulation by $G\alpha_i$-$G\gamma$ dissociation assay (left column) or TUG891 stimulation by $G\alpha_q$-$G\gamma$ dissociation assay (right column). The effects of mutations on EC50 changes in responses to 9-HSA or TUG891 for corresponding pathways were indicated by the red color scale. The heatmap are generated according to the data shown in fig. S27, B and C (refer to table S17 for details). (**J**) Heatmap representation of the effect of mutations in the GPR120 interface on G_i or G_s coupling in response to 9-HSA stimulation by $G\alpha_i$-$G\gamma$ dissociation assay (left column) or EPA stimulation by $G\alpha_s$-$G\gamma$ dissociation assay (right column). The effects of mutations on EC50 changes in responses to 9-HSA or EPA for corresponding pathways were indicated by the blue color scale. The heatmap are generated according to the data shown in fig. S27, B and D (refer to table S17 for details). (**K**) Structural representation showing no interactions between E135$^{3.49}$/Y227$^{5.58}$ and C351$^{G.H5.23}$ in 9-HSA-GPR120-G_i complex. (**L**) Structural representation of the interaction of E135$^{3.49}$ and Y227$^{5.58}$ with Y391$^{G.H5.23}$ in GPR120-G_s according to MD simulation. The blue dotted line represents a hydrogen bond. (**M**) Structural representation showing no interactions between Q144^{ICL2} and I344$^{G.H5.16}$ in 9-HSA -GPR120-G_i complex. (**N**) Structural representation of the interaction of Q144^{ICL2} with Q384$^{G.H5.16}$ in GPR120-G_s according to MD simulation. The red dotted line represents a potential hydrogen bond.



A
GPR120-Gq path
GPR120
Ligand
F311 7.43
F88 2.53
I89 2.54
L86 2.51
N58 1.50
D85 2.50
F82 2.47
N317 7.49
L81 2.46
V78 2.43
Y321 7.53
Y227 5.58
R136 3.50
Gαq
B
GPR120-Gi
GPR120-Giq
F311 7.43
F88 2.53
I89 2.54
L86 2.51
D85 2.50
N313 7.45
N58 1.50
N317 7.49
F82 2.47
C
L81 2.46
F82 2.47
Y321 7.53
V78 2.43
R136 3.50
Y227 5.58
Y356
D
Gq
Gi
WT
D85A
F88A
I89A
Y227A
F311A
N317A
Y321A
EC50 fold change .vs. WT
E
GPR120-Gs path
GPR120
Ligand
F211 5.42
N215 5.46
S123 3.37
Y165 4.52
L127 3.41
I162 4.49
L158 4.45
L77 2.42
E135 3.49
Gαs
F
GPR120-Gi
GPR120-Gs
S123 3.37
F211 5.42
N215 5.46
Y165 4.52
G
Y165 4.52
L127 3.41
I162 4.49
L77 2.42
L158 4.45
Y391
E135 3.49
H
Gs
Gi
WT
L77A
L127A
E135A
L158A
I162A
Y165A
EC50 fold change .vs. WT
I
LA
R254 6.25
LA-GPR120-Gi
J
TM6
R254 6.25
K314
E308
LA-GPR120-Gi
K
LA
EPA
ALA
DHA
Gi
Gs
Gq
5≤
EC50 fold change

Fig. 7. The propagating paths of GPR120-G_s/G_q and SNP in GPR120. (**A**) Schematic diagram represented the propagating paths that contribute to the G_q vs. G_i bias decision mechanism by comparing the structures of 9-HSA-GPR120-G_i and TUG891-GPR120-G_q complexes. The propagating paths started with the engagement with the ligand binding pockets and extended to the G_q protein interface region. The green cartoon represents GPR120. The yellow cartoon represents $G\alpha_q$. The bold purple curve represents the interface region between GPR120 and G_q. (**B** and **C**) The structural rearrangements of key residues along the propagating paths connected the ligand binding pocket to cytoplasmic side by comparison of 9-HSA-GPR120-G_i complex and TUG891-GPR120-G_{iq} complex structures. Hydrogen bonds or polar interactions are depicted as blue dashed lines. The hydrogen bond formed between N317 and N313 in 9-HSA-GPR120-G_i structure are highlighted as a red dashed line. (**D**) Heatmap representation of the effect of mutations in the propagating paths of GPR120-G_q or GPR120-G_i in response to TUG891 stimulation by $G\alpha_q$-$G\gamma$ and $G\alpha_i$-$G\gamma$ dissociation assay. The changes of EC50 in corresponding pathways were indicated by the purple color scale. The heatmap is generated according to the data shown in figs. S22, B and C, and S29, D and E (refer to table S20 for details). (**E**) Schematic diagram represented the propagating paths that contribute to the G_s vs. G_i bias decision mechanism by comparing the structures of 9-HSA-GPR120-G_i complex and simulated EPA-GPR120-G_s complex. The propagating paths started with the engagement with the ligand binding pockets and extended to the G_s protein interface region. The green cartoon represents GPR120. The orange cartoon represents $G\alpha_s$. The bold pink curve represents the interface region between GPR120 and G_s. (**F** and **G**) The structural rearrangements of key residues along the propagating paths connected the ligand binding pocket to cytoplasmic side by comparison of 9-HSA-GPR120-G_i complex and simulated EPA-GPR120-G_s complex structures. Hydrogen bonds are depicted as blue dashed lines. (**H**) Heatmap representation of the effect of mutations in the propagating paths of GPR120-G_s or GPR120-G_i in response to EPA stimulation by $G\alpha_s$-$G\gamma$ and $G\alpha_i$-$G\gamma$ dissociation assay. The changes of EC50 in corresponding pathways were indicated by the orange color scale. The heatmap is generated according to the data shown in figs. S22, E and F, and S29, F and G (refer to table S22 for details). (**I**) Structural representation of the position of GPR120 SNPs R254$^{6.25}$ (R270H in the long isoform of GPR120) in the LA-GPR120-G_i complex structure model, which located in the TM6 of the receptor and participated in the GPR120-G_i trimer interface. Red ball: the site R254$^{6.25}$. (**J**) Detailed interactions of R254$^{6.25}$ of GPR120 with G_i, by forming contacts with the side chains of E308 and main chain carbonyl group of K314 via charge-charge or polar interactions, respectively. The charge-charge and polar interactions were depicted as red dashed lines. (**K**) Heatmap representation of R254H mutation effects on ligands (LA, EPA, ALA and DHA) induced GPR120 activities by different G protein dissociation assays. Notably, R254H mutation significantly reduced the G_i, G_q and G_s coupling of GPR120 in response to stimulations of these FAs. Cross indicates that no G_s activity could be detected of GPR120 in response to stimulation with LA. The heatmap is generated according to the data shown in fig. S30, G to Q (refer to table S24 for details).